

Can return-chasing explain the failure of Uncovered Equity Parity in emerging markets?

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Abstract

Within a portfolio-rebalancing framework, the Uncovered Equity Parity (UEP) predicts that countries with strong equity markets should experience a currency depreciation, as higher total returns in domestic equity market will cause foreign investors to repatriate some of their investments to decrease their exchange rate exposure, leading to exchange rate depreciation. Using daily data including all the recorded trades of foreign equity investors for six Asian Emerging Markets from the 1990s to 2013, we find a positive rather than a negative relationship between currency and equity returns. We document that it is because the foreigners in aggregate chase returns rather than rebalance their portfolios in emerging markets, while foreign equity flows do cause exchange rate movements in the same direction. Additionally, we find little evidence that foreign equity flows respond to past currency returns, suggesting that foreign equity investors only use local currency as a vehicle investing in emerging markets. [Word count: 149]

Keywords: Uncovered Equity Parity; equity flows; equity returns; foreign exchange rate; portfolio-rebalancing; return-chasing; vehicle currency.

JEL classification: F31; G10; G11; G15

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“The increasing size and equity content of current capital flows has not yet inspired a new financial market paradigm for exchange rate theory, in which exchange rates, equity market returns, and capital flows are jointly determined.” (Hau and Rey, 2006)

1. Introduction

A relatively recent parity condition, the uncovered equity parity (UEP) condition, has been proposed in the international finance literature by Hau and Rey (2004, 2006). The main intuition behind UEP is one of portfolio-rebalancing (Hau and Rey, 2006, p277). Under the assumption of incomplete exchange rate (hereafter FX) risk hedging¹, UEP has two steps and one prediction: First, when the total returns of domestic equity holdings outperform foreign holdings (due to shocks from either equity or FX markets), foreign investors are exposed to higher relative FX exposure and decide to repatriate some of the domestic equity to decrease the FX risk. Second, the associated selling of domestic currency leads to domestic currency depreciation. Therefore, UEP predicts a (theoretically perfect) negative correlation between the performance of FX and equity returns in local currency.

UEP is of essential importance for at least two reasons. On the one hand, it asserts that foreign equity flows can explain FX movements, which have been notoriously difficult to predict using other macro-economic variable (for a seminal paper, see, Meese and Rogoff, 1983). On the other hand, from the perspective of international investing and portfolio management, it is also important for global investors, as in most cases investments in equity markets of different countries inevitably involve investments in corresponding FX markets.

Albeit of importance, only a few research papers empirically test UEP, perhaps because of the limited data availability of international capital flows². While Curcucu et al. (2014) investigate the two steps of UEP but not its prediction, the other papers mostly abstract from capital flows and only examine its prediction, the negative correlation between the FX returns and equity return differentials, or the correlation between FX returns and domestic equity returns in local currency (hereafter equity local returns). For instance, Hau and Rey (2006) propose UEP and show a negative correlation between equity return differentials and FX returns within a sample of 17 OECD countries vis-à-vis the US. Similar to UEP, Cappiello and

¹ Cenedese et al. (2014) provide an alternative derivation of UEP, which does not rely on this imperfect hedging assumption, but uses absence of arbitrage and standard asset pricing theory.

² For instance, Cho et al. (2014) note “Unfortunately, testing these conjectures empirically is not easy, mainly because of the lack of appropriate data”.

De Santis (2007) propose a negative correlation between expected equity return differentials and expected FX returns and verify it using monthly data from the UK, Germany and Switzerland vis-à-vis the US. However, the developed economies and the emerging markets should be treated separately (Kim, 2011). Kim (2011) finds a positive correlation between equity local returns and FX returns in four Asian emerging markets (hereafter EMs), and terms it as the failure of UEP in EMs. The failure of UEP in EMs has been confirmed by Cenedese et al. (2014)³, who use a portfolio approach and find that on average FX movements do not offset equity return differentials in a cross-section of 43 countries (including both developed countries and EMs).

However, it may not be innocuous to abstract from capital flows and hold the sign (and/or the magnitude) of the correlation between the FX returns and equity return differentials (or the correlation between FX returns and equity local returns) as the indicator of the validity of UEP. The key driver of UEP is portfolio-rebalancing, which can be a strategy adopted for a subset of foreign investors but may not for all foreign investors in aggregate in EMs. As a result, UEP may hold for the specific subset of foreign investors adopting the strategy of portfolio-rebalancing but not for all foreign investors in aggregate in EMs. In fact, the capital flow literature (see, e.g., Griffin et al., 2004; Richards, 2005) suggests that international equity flows respond negatively to the *push factors* such as equity returns in developed countries (portfolio-rebalancing), but respond positively to the *pull factors* such as equity returns in EMs (return-chasing). If the foreign investors in aggregate chase equity returns in EMs, it should not be surprising to find that the FX and equity local returns are positively correlated in EMs.

So far, we are aware of only a couple of research papers, which have empirically tested UEP with capital flow data after Hau and Rey (2004, 2006). The first one is Curcuro et al. (2014), who test the two steps of UEP separately with data on U.S. investors' monthly equity positions across 42 markets from 1990 to 2010. Curcuro et al. (2014) cannot test the prediction of UEP on the correlation between FX returns and equity local returns, as it can only be tested with equity flow data of all foreign investors, rather than any subset of foreign investors in the

³ In the appendix section, Cenedese et al. (2014) find a negative correlation between equity local returns and FX returns for almost every developed country but a positive correlation for almost every EM.

country. The rationale is that the effects on FX markets of any subset of foreign investors may be offset by other subsets of foreign investors if they trade against each other. The other one is Cho et al. (2014), who construct quarterly net capital flow data for 9 developed and 12 emerging markets from 1996 to 2009 from Balance of Payments account data reported by the International Monetary Fund (IMF). The data set used in Cho et al. (2014) covers all foreign investors, but as they admitted “the data employed in this analysis are somewhat crude in terms of its frequency”. Since equity and FX markets can fluctuate substantially even within a day, the problem of information loss can be really serious if the matched capital flow data are in low frequency such as monthly or quarterly. Accordingly, in this paper we examine the two steps and the prediction of UEP using daily equity flow data including all the recorded trades of foreign investors for six Asian EMs from the 1990s to 2013.

To the best of our knowledge, we are the first to examine both the two steps and the prediction of UEP. In line with the traditional literature on the relationship between international capital flows and asset returns, the main methodology in this paper is vector autoregressive models, including both reduced-form vector autoregressive models (VAR) and structural vector autoregressive models (SVAR). We start by probing the contemporaneous relationship among flows, FX returns and equity local returns in our preliminary data analysis. We confirm a positive correlation between FX and equity local returns in EMs found in a few recent papers (Kim, 2011; Cho et al., 2014; Cenedese et al., 2014). Since the failure of UEP in EMs may arise from either the first step or the second step of UEP, we tackle both steps in turn.

The first step of UEP states that foreign equity investors rebalance away from (toward) countries whose equity/FX markets have recently performed well (poorly), but we find contradicting and asymmetric results in domestic equity and FX markets. On the one hand, foreign equity flows respond positively to the past equity local returns. On the other hand, foreign equity flows are insensitive to the past FX returns, echoing with the results of Curcuru et al. (2014) for U.S. equity investors. Thus, the hypothesis of portfolio-rebalancing is clearly rejected in both domestic equity and FX markets in our sample of six EMs.

Then we assess the motives behind the responses of foreign equity investors to past equity local returns by decomposing the current returns into an expected component and an

unexpected component, and find that the responses of foreign equity investors are mainly due to the expected equity local returns. In other words, foreign equity investors chase expected equity local returns, and past equity returns signal expected current equity returns because of the momentum in equity returns. This would be consistent with the little evidence that foreign equity investors respond to past FX returns, as there is little momentum in FX returns.

As regards to the second step of UEP, we find a strong contemporaneous positive relationship between foreign equity flows and FX returns, and a weak inter-temporal relationship which implies that foreign equity flows may also have a positive and permanent impact on future FX returns.

This paper contributes to the literature in several directions. First of all, we unveil another side of UEP due to a different mechanism in the first step, and find that UEP does not automatically guarantee a negative correlation between domestic equity and FX returns. The relationship between FX and equity local returns hinges on the overall behavior of foreign equity investors in aggregate, i.e., whether they pursue a portfolio-rebalancing or return-chasing strategy. The former predicts a negative correlation, while the latter creates a positive correlation between FX and equity local returns. In other words, we reconcile the mixed evidence on the prediction of UEP about the correlation between FX and equity local returns in developed and emerging markets.

Second, we find distinct mechanism in FX markets from equity markets in the first step of UEP, as we find litter evidence that foreign equity flows respond to past FX movements, suggesting that foreign investors in aggregate in EMs mainly use exchange rates as a vehicle (Goldberg and Tile, 2008; Devereux, and Shi, 2013). It is consistent with the vital assumption of incomplete hedging of FX risk in UEP, and the extremely low hedge ratios for foreign equity investment (e.g., Levich et al., 1999; Curcuru et al., 2014). Given the huge volatility in FX markets even at short horizons (see, e.g., Bank of International Settlements, 2013), it would be surprising to find that foreign equity investors systematically respond to past FX movements.

Third, we provide additional evidence on the second step of UEP, as we find a strong contemporaneous positive relationship between foreign equity flows and FX returns, and a weak inter-temporal relationship which implies that foreign equity flows may also have some

positive and permanent impact on future FX returns. Permanent impacts on FX returns are usually due to private information incorporated in currency order flows⁴ (e.g., Evans and Lyons, 2002a, b, c). Given the difficulty in forecasting FX dynamics at short horizons (e.g., Meese and Rogoff, 1983) and the inactive trading pattern of foreign equity investors (Richards, 2005, p5), it is hard to imagine that foreign equity investors hold private information about FX markets. A more likely explanation is that the net foreign equity flows and the currency order flows are closely aligned as documented in UEP⁵. The only difference between our analysis and the one in Hau and Rey (2004, 2006) is that we find it in a return-chasing rather than portfolio-rebalancing framework.

Finally, we contribute to another unsettled debate whether foreign investors pursue a return-chasing or a portfolio-rebalancing strategy regarding equity local returns (e.g., Curcucu et al., 2011). Previous results either come from low frequency such as monthly data (e.g., Curcucu et al., 2011), or from a short span of daily data one decade ago (e.g., Griffin et al., 2004; Richards, 2005). We find that foreign equity investors in aggregate pursue a return-chasing rather than portfolio-rebalancing strategy regarding equity local returns in EMs, in consistence with most of the studies (e.g., Griffin et al., 2004; Richards, 2005). However, we do not exclude the possibility that a specific group of investors such as the U.S. equity investors in Curcucu et al. (2011) might pursue a portfolio-rebalancing strategy.

The rest of the paper unfolds as follows. In the next section, we discuss the related literature. Section 3 outlines the data and preliminary data analysis. Section 4 examines the first and the second step of UEP in turn. Section 5 compares our explanation with the other existing explanations in literature, while Section 6 checks robustness. Section 7 concludes with a summary.

⁴Although net capital flows and order flows are similar in nature, they are completely two things. While net capital flows is the net of foreigners' net purchases from the residents and the residents' net purchases from foreigners in domestic markets, order flows is the "net of buyer-initiated and seller initiated orders"(Evans and Lyons, 2002a). There are at least two kinds of order flows in the existing literature: currency order flows, equity order flows.

⁵Hau and Rey (2004, p127) note that "Yet simple portfolio shifts could also give rise to order flow without any role for information asymmetries. Within the portfolio-rebalancing framework and conditional on exogenous equity return and exchange-rate shocks, it is plausible that net capital flows and order flows are closely aligned. Conditional on an exogenous appreciation of his foreign wealth for example, the home investor is likely to initiate the selling of foreign assets as well as the selling of foreign currency balances."

2. Literature review

Corresponding to the two steps of UEP, our paper mainly relates to two strands of international finance literature dealing, with the reactions of foreign investors to past equity local returns and FX returns, and with the impacts of foreign equity flows on FX returns, respectively.

We start with the strand of literature on the reactions of foreign investors to past equity local returns, which is still unsettled. In a seminal paper, Bohn and Tesar (1996) use an intertemporal international capital-asset-pricing model to decompose the net purchases of U.S. equity investors in other markets into: 1) transactions that are necessary to maintain a balanced portfolio of securities, so-called “portfolio-rebalancing”, and 2) net purchases that are triggered by time-varying investment opportunities. They find U.S. transactions in other markets are primarily driven by the latter effect, as U.S. investors tend to move into markets where returns are expected to be high and retreat from markets when expected returns are low, so-called “return-chasing”. Since then, return-chasing has been seen as a stylized fact, confirmed by subsequent studies (Froot et al., 2001; Griffin et al., 2004; Richards, 2005; Froot and Ramadorai, 2008) and incorporated in theoretical models (Brennan and Cao, 1997; Guidolin, 2005; Albuquerque et al., 2007, 2009; Dumas et al., 2014). However, as criticized by Curcuru et al. (2011), most of the previous empirical studies⁶ use bilateral flow data and cannot perfectly control for the effects from the changes in financial wealth of the investors. Using monthly portfolio holdings data and portfolio-based techniques, Curcuru et al. (2011) find that U.S. equity investors neither chase equity returns nor buy past losers, but sell past winners, a form of partial rebalancing.

With less literature about the effects of past FX returns on flows, the case here is no clearer at all. For instance, Adler and Dumas (1983) suggest that foreign equity investors will hedge their equity purchases against currency risk and take no actions regarding past FX movements. More importantly, in this case foreign equity investors have no essential impact on FX markets and UEP degenerates into the simple interactions between equity local returns

⁶Two noteworthy exceptions are Froot et al. (2001) and Froot and Ramadorai (2008), who use daily portfolio holdings data over the period from 1994 to 1998 from State Street Company and find that US equity investors chase equity local returns as well.

and equity flows without a role of FX (Froot et al., 2001; Griffin et al., 2004; Richards, 2005). However, surveys of investors suggest that international equity positions are typically unhedged (e.g., only 8% according to Levich et al., 1999). This is not only true in national statistics but also at the level of individual equities (Curcuru et al., 2014, p90). Hau and Rey (2004, 2006) suggest that foreign equity investors will repatriate some of the investment when FX appreciates, in stark contrast with the “currency carry trade” literature (Lustig and Verdelhan, 2007; Burnside et al., 2008; Brunnermeier et al., 2008; Lustig et al., 2011), which states that investors should increase their allocations to the currencies that have performed well and these currencies would continue to appreciate. Menkhoff et al. (2014) find that FX investors can either be positive feedback investors or negative feedback investors via currency order flows. Curcuru et al. (2014) find evidence that U.S. equity investors do not react to currency movements.

There are few research papers analyzing the impact of foreign equity flows on FX returns, although many about the impacts of foreign equity flows on equity local returns (e.g., Froot et al., 2001; Griffin et al., 2004; Richards, 2005; Froot and Ramadorai, 2008). Perhaps the only one is Hau et al. (2010), who find a downward sloping demand curve in FX markets and a FX impact of the equity flows arising from the redefinition of the Morgan Stanley Capital International (MSCI) Global Equity Index in 2001 and 2002. While Froot and Ramadorai (2005) find currency flows of intuitional investors can only cause short-term price pressures in FX markets, studies on microstructure generally suggest that currency order flows can cause *permanent* rather than *temporary* FX returns due to incorporated private information (e.g., Evans and Lyons 2002a, b, c; Hau et al. 2002; Killeen et al. 2006). Bridging the studies on macroeconomic and microstructure studies, Hau and Rey (2004) explicitly point out that, as a part of UEP, net foreign equity flows and currency order flows are closely aligned. While Hau and Rey (2006) and Curcuru et al. (2014) find evidence of a positive contemporaneous correlation between foreign flows and FX returns, foreign equity flows may also have a *permanent* impact on future FX returns in the same direction (Hau and Rey, 2004). It is difficult to detect this inter-temporal relationship within low-frequency data, while our daily data put us in a better position to investigate the potential effects of past foreign equity flows on FX returns.

3. Data

In this section, we report the source, description, the comparison of our data with data sets used in previous literature, and the preliminary analysis of our data sets.

3.1 Description of data

Our data set mainly consists of net equity flows, FX returns, and equity local returns in daily frequency for six Asian EMs.⁷ Our capital flow data set has two main advantages, which makes it an ideal candidate to test UEP. On the one hand, since the data set includes all the recorded trades of foreign investors from the stock exchanges, it has a broader coverage than data covering only one group of investors – for example, U.S. investors in studies using data from U.S. Treasury (e.g., Brennan and Cao, 1997; Hau and Rey, 2004, 2006) or Federal Reserve (Curcucu et al., 2011, 2014), or mutual funds (e.g., Hau and Rey, 2008) or customers of a particular custodian (Froot et al., 2001, Froot and Ramadorai, 2005, 2008). On the other hand, daily data allow a “precise” analysis of the short-term effects and determinants of foreign flows (Richards, 2005, p.7)⁸. If foreign equity flows cause returns or respond systematically to recent returns in FX or domestic equity markets, these linkages should be captured in our data.

Our data of daily net equity flows in the six East Asian markets are obtained via the exchanges of the markets, Bloomberg and CEIC databases.⁹ A small number of obvious errors have been observed by cross-checking different databases. We drop a small percentage of unreliable earlier sample and the winsorize the data at 99% level. The final sample begins from September 9, 1996 for Indonesia (JSX), June 30, 1997 for Korea (Kospi), March 15, 1999 for

⁷ As claimed by Richards (2005): “The sample size of six markets is large enough to provide results that are potentially fairly general, yet is small enough to allow more attention to market-specific data and modeling issues than might be possible in datasets with a larger number of markets.”

⁸ In the terminology of Richards (2005), this data set is “precise” as it records the actual trade dates. Proprietary data for flows such as the one used in Froot, et al. (2001) are based on contractual settlement dates, and the trade dates are inferred from settlement conventions in each country.

⁹ Similar data sets have been used in Richards (2005) or Griffins et al, (2004) but with a much shorter span (around three years). Details of CEIC database can be found as follows (<http://www.ceicdata.com>). Richards (2005) provides a detailed description of the data of capital flows.

Korea (Kosdaq)¹⁰ and Philippines (PSE), January 1, 2001 for Taiwan (TWSE)¹¹, and December 1, 1997 for Thailand (SET). The ending date is December 30, 2013 for all markets. We also obtain the daily market capitalization of each market from Bloomberg and scale daily net purchases of foreigners by local market capitalization so that the scaled flows we actually use are in percentages.^{12, 13}

Equity local returns (in %) are constructed as “log returns” of the main capitalization-weighted index of stocks traded on these markets in local currency. Ideally, UEP should be tested with the time-varying holding weight of each individual local stock for every foreign investor, so that researchers can calculate the portfolio returns earned by all foreign investors in aggregate. While the directly measured returns series based on foreign investors' holdings do not exist, the literature suggests using publicly available country-level equity indices comprising of the largest and most liquid firms in each country, as foreigners tend to hold the largest and most liquid domestic stocks (see Curcuru et al., 2014 and the relevant references therein). We use the daily closing prices of Jakarta JSX Composite Index in Indonesia, the Kospi Index and the Kosdaq Index in Korea, the PSE Composite Index in the Philippines, the TWSE/TAIEX Index in Taiwan, and the Bangkok SET Index in Thailand, respectively. Unlike some of other indices provided by international providers such as Morgan Stanley Capital International (MSCI) and Standard and Poor's (S&P) indices, these indices are actually the “headline” indices available to investors on a real-time basis.

¹⁰ The first five are traditional “main boards”, while Kosdaq is a “second board” focusing on start-up and technology-related companies in Korea. Both the first five markets and Kosdaq have been studied in Richards (2005) as well. “Foreign investors in these markets must register with the local exchange or regulator, and brokers must report the nationality of the buyer and seller in each transaction that occurs. The resulting data capture the trading of all registered foreign investors (Richards, 2005)”.

¹¹ For Taiwan (TWSE), we have data from Oct 25, 2000 but only used data from January 1, 2001 due to two reasons. On the one hand, there is Saturday trading in Taiwan on the first, third and fifth Saturdays of each month in 2000. On the other hand, the 75% foreign investment ownership limit has been removed at the end of 2000.

¹² Following Froot et al. (2001) and Richards (2005), we do not include net purchases by foreigners of American Depositary Receipts (ADRs) or country funds in foreign markets, equity futures or other derivatives in the domestic markets. Curcuru et al. (2014) provide a snapshot of the end-2010 amounts of the international positions of U.S. investors, and conclude that it is impossible to tell the real currency exposure faced by the U.S. investors with publicly available data. The data problems with the derivatives are more daunting.

¹³ In daily frequency and over such a long span, we are only able to obtain data for the six markets which have floating exchange rates. We are aware that Ulku and Weber (2014) use data from May 4, 2004 to April, 30 2012 for a European country — Turkey from the Central Registry Agency and Clearing and Custody Bank of Turkey. Unfortunately we find difficulties in accessing that data. Yet, the regional movement of capital flows in Europe and Asia may not be the same.

Daily FX returns (in %) are constructed as the “negative log returns” of the daily exchange spot rate data. The conventional market quotation is the number of local currency per U.S. dollar, and positive FX returns mean local currency appreciation. We are aware that local currency can be priced by currencies other than the U.S. dollar (e.g., Cho et al., 2014), but the case should be similar due to the famous triangle arbitrage in FX markets. As a result, the local currency is priced by the U.S. dollar all over this paper. Importantly, the exchange rates are neither under fixed nor managed float exchange rate system over our sample period for these countries. We choose the spot exchange rates exactly corresponding to the closing time of domestic equity markets from the WMR/Reuters database via DataStream, Bloomberg and local exchanges.

In comparison with the data sets used in previous literature, we employ a relatively high frequency (daily) and long span (more than one decade) data set including all the recorded trades of foreign investors for six Asian emerging equity markets, allowing a very precise examination of UEP. In contrast, many previous papers use monthly FX and equity returns data without considering capital flows, such as Cappiello and De Santis (2007), Kim (2011) and Cenedese et al. (2014). Only a few papers have tried the capital flow data, and the data in these papers are less appealing in terms of testing UEP than our data.

For instance, after introducing their equity and FX data, Hau and Rey (2006, p298) note that “portfolio flow data are more difficult to obtain”. Hau and Rey (2004, 2006) use monthly bilateral equity flows between the U.S. and OECD developed countries from the U.S. Treasury International Capital (TIC) database, acknowledging the famous shortcoming that equity transactions in TIC database are recorded by the nationality of the person with whom the transaction is carried, not by the country that originally issued the security (Hau and Rey, 2006, p299).

Cho et al. (2014) try to explain the magnitude of the correlation between FX and equity local returns by using quarterly Balance of Payment data from the International Financial Statistics (IFS) reported by the IMF to construct net capital flows. Cho, et al. (2014) explicitly note *“Since we are using data over quarterly intervals, information loss would be more serious than when we use finer data, for example, over monthly intervals. Not only the number of*

observations is reduced, but also inter-temporal changes in variables within the quarter are netted out, making the power of statistical tests smaller. Therefore, we conjecture that if we are able to use data at higher frequency, we might be able to obtain more significant results”.

3.2 Preliminary data analysis

The properties of the three variables that are the focus of our analysis – net flows (NF_{it}), equity local returns (ELR_{it}) and FX returns (FXR_{it}), – are shown in Table I over the period from various starting dates in the 1990s to December 30, 2013, since the sample period differs for each market.

We report the starting date of the sample, the mean, median, and standard deviation of net flows and returns, the first five autocorrelation estimates for each series, and the contemporaneous correlation coefficients between net flows and equity local returns, equity local returns and FX returns, and net flows and FX returns, respectively.

The mean values of net flows (NF_{it}) are all positive, indicating that over the whole sample there are more purchases than sales by foreign equity investors in these six EMs. In contrast, the mean values of FX returns (FXR_{it}) and equity local returns (ELR_{it}) can be both positive and negative, which demonstrates the heterogeneity of our data. However, the median values are all positive, implying the profitability of investing in EMs. The standard deviation of net flows (NF_{it}) varies across markets, from 0.013% for Philippines (PSE) to three times as much (0.040%) for Taiwan (TWSE), which is consistent with Griffin et al. (2004) to find that in all markets most daily foreign net activity is generally less than 0.1% of market capitalization.

Consistent with previous literature on capital flows (Froot et al., 2001; Griffin et al., 2004; and Richards, 2005), we find substantial positive autocorrelation in daily net flows (NF_{it}), with a median first-order autocorrelation coefficient of 0.451. The autocorrelation in net flows declines slowly and is significant over the past 5 lags in general, perhaps due to motives to mitigate market impacts or the heterogeneous information processing speeds of different types of investors (Griffin et al., 2004). Daily returns in equity markets are also significantly autocorrelated with its first lag, with a median first-order autocorrelation coefficient of 0.100 for equity local returns (ELR_{it}). In contrast, daily FX returns (FXR_{it}) are not correlated with its

first lag except for Thailand (SET), with a median first-order autocorrelation coefficient of -0.002. Unlike equity local returns, there is little momentum in FX returns, and previous day's FX returns provide little information for the current FX returns.

The last three columns show contemporaneous correlation between net flows and equity local returns, equity local returns and FX returns, and net flows and FX returns, respectively. While UEP suggests that the correlation coefficient between past equity local returns and current foreign equity flows should be negative, we find a positive contemporaneous correlation between equity local returns and flows. In particular, within each market there is a strong positive same-day correlation between daily net flows and equity local returns, with a median correlation coefficient of 0.304, consistent with previous literature on capital flows and equity local returns (e.g., Froot et al., 2001; Griffin et al., 2004; Richards, 2005). Interestingly, for every market there is also a statistically significant positive correlation between equity local returns and FX returns ranging from 0.20 (Indonesia and Thailand) to 0.31 (Taiwan), with a median correlation coefficient of 0.269. The prediction of a negative correlation between equity local returns and FX returns is clearly rejected here, and the size of the correlations is comparable to the results from previous literature (e.g., the size of pairwise correlation between weekly equity local returns and FX returns varies from 0.121 (Czech) to 0.485 (Mexico) in EMs in panel B of Table III in Cho et al., 2014). Although it is unclear whether the failure of UEP in EMs arises from its first step (portfolio-rebalancing) or its second step (equity flows cause FX returns), we also find a statistically significant positive same-day correlation between daily net flows and FX returns with a median correlation coefficient of 0.103, providing some preliminary support for the second step of UEP.

4. Empirical assessment of UEP in EMs

In this section, we examine the two steps of UEP in turns. We start by examining whether foreign equity investors pursue a portfolio-rebalancing strategy regarding the past equity local returns and past FX returns as suggested by the first step of UEP. After that, we examine the motives behind the behaviors of foreign equity investors, i.e., do foreign equity investors chase returns in EMs indeed? Finally, we examine whether foreign equity flows cause FX returns as

suggested by the second step of UEP.

4.1 The first step of UEP: do foreign equity investors rebalance?

The first step of UEP indicates that foreign equity investors in aggregate pursue a portfolio-rebalancing strategy regarding the past total equity returns (including equity local returns and FX returns) in EMs, which itself has two parts. One part is that foreign equity investors pursue a portfolio-rebalancing strategy regarding the past equity local returns in EMs. The second part is that foreign equity investors also rebalance from the FX returns in EMs. Curcuru et al. (2014) find no support for the second part but strong support for the first part, which cannot explain the failure of UEP in EMs. We examine these two parts step-by-step.

In line with the previous literature on international capital flows and domestic equity returns such as Froot et al. (2001), Griffin et al. (2004) and Richards (2005), we utilize a structural vector autoregressive models (SVAR) with 5 lags (motivated below) for each country where we cast the joint dynamics of the variables for *Net Flows* (NF_{it}) and *Equity Local Returns* (ELR_{it}) in model (1) below. The VAR is structural as we include contemporaneous equity local returns into the flows equation (1b), which is a key setup to test the first step of UEP. We are particularly interested in whether the past equity local returns positively or negatively predict flows over and above the predictions of lagged flows, after controlling for the contemporaneous equity local returns.¹⁴

$$ELR_{i,t} = \alpha_{ELR} + \sum_{d=1}^5 \phi_{1,1,d} * ELR_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * NF_{i,t-d} + u_{i,t}^{ELR} \quad (1a)$$

$$NF_{i,t} = \alpha_{NF} + \sum_{d=0}^5 \phi_{2,1,d} * ELR_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * NF_{i,t-d} + u_{i,t}^{NF} \quad (1b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{ELR} \\ u_{i,t}^{NF} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{ELR,NF} \right], \mathbf{D}_i^{ELR,NF} = \begin{bmatrix} \sigma_{u,i,ELR}^2 & 0 \\ 0 & \sigma_{u,i,NF}^2 \end{bmatrix}$$

¹⁴ The SVAR here only allows for a contemporaneous effect of returns on flows, which is different from the typical SVAR with only the contemporaneous flows in the returns equation, stemming from Hasbrouck (1991). We ask how the past returns affect flows, while most literature focuses on how the past flows affect returns. As pointed out by Ulku and Weber (2013, p.2734), while the set-up in Hasbrouck (1991) may be legitimate under a dealer system without frictions with tick data, flows may also be affected by contemporaneous returns with daily or less frequent data due to intra-period feedback trading (Brenan and Cao, 1997). Similar to Ben-Rephael, et al. (2011), we use the typical set-up of Hasbrouck (1991) later for completeness and find similar results.

However, there is a risk of missing variables associated with this set-up. To the extent that FX returns are contemporaneously correlated with equity local returns, as we have shown in the preliminary analysis, a positive or negative relationship between foreign flows and domestic equity (FX) returns could simply be proxying for a FX (equity) effect. (Griffin et al., 2004, p652). This conjecture is plausible but there is little evidence about it. To test this conjecture, we follow Griffin et al. (2004, p652) and add five lags of FX returns as an exogenous variable into the SVAR and compare the results from the model (2) below with the ones from model (1):

$$ELR_{i,t} = \alpha_{ELR} + \sum_{d=1}^5 \phi_{1,1,d} * ELR_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,3,d} * FXR_{i,t-d} + u_{i,t}^{ELR} \quad (2a)$$

$$NF_{i,t} = \alpha_{NF} + \sum_{d=0}^5 \phi_{2,1,d} * ELR_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,3,d} * FXR_{i,t-d} + u_{i,t}^{NF} \quad (2b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{ELR} \\ u_{i,t}^{NF} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{ELR,NF} \right], \mathbf{D}_i^{ELR,NF} = \begin{bmatrix} \sigma_{u,i,ELR}^2 & 0 \\ 0 & \sigma_{u,i,NF}^2 \end{bmatrix}$$

We order the equity local returns before flows, so that we can make sure that flows do not affect equity local returns contemporaneously through a simple Choleski factorization. No matter whether including an exogenous variable or not, our SVAR systems are exact identified and can be estimated separately for each country as seeming unrelated regressions (SURs). Before presenting the empirical results from various models, we check that all eigenvalues having moduli less than one so that our SVARs are stationary. Unlike Froot et al. (2001), we do not restrict the autoregressive coefficients to be the same across countries, as the degree of freedom is not a problem for us. By the Hannan–Quinn Information Criterion, the lag length is suggested to be from 2 to 5, and 5 in most of cases. As a result, we set the lag length at 5 as Griffin et al. (2004) and Richards (2005), which means that we are examining weekly effects with daily data as five trading days forming one week.

Table II shows the results from the equity local returns equation for the bivariate SVAR of foreign equity flows and equity local returns with no exogenous variable in Panel A, and for the bivariate SVAR of foreign equity flows and equity local returns including FX returns as an exogenous variable in Panel B, respectively. We report the estimates of the contemporaneous

net flows and the past net flows up to 5 lags, adjusted R^2 and the p-value of Granger causality test for the hypothesis: Past equity local returns do not Granger-cause flows. In general, the results from Panel A and B are similar.

The results in both Panel A and B of Table II show that past equity local returns positively predict flows over and above the predictions of lagged flows, as the sum of the coefficients of past equity local returns is positive, after controlling for the contemporaneous equity local returns. In fact, the estimated coefficients of one-day lagged equity local returns are all positive and strongly significant. For instance, in both Panel A and Panel B of Table II, the estimated coefficients of one-day lagged equity local returns are highly significant, ranging from 0.001 (PSE) to 0.005 (Kospi). In all six markets, Granger causality tests strongly reject the null hypothesis that past equity local returns do not Granger-cause flows with p-values less than 0.001. The adjusted R^2 ranges from 0.105 (PSE) to 0.516 (TWSE) in Panel A, and 0.104 (PSE) to 0.516 (TWSE) in Panel B, which is comparable to the existing literature on the interaction of foreign equity flows and equity local returns (Griffin et al., 2004; Richards, 2005).

Figure I presents the responses of flows to a one-standard deviation innovation in past equity local returns using general impulse response functions (GIRFs). We only report the results based on equation (2b), as the results based on equation (1b) are similar. The GIRFs are invariant to the ordering of the variables in the VARX so that there is no need of assumptions on the sequence of shocks (Pesaran and Shin, 1998). Over a 10-day period, we find that the current equity local returns have a positive and significant influence even on the next day's foreign equity flows in all 6 sample markets, which is consistent with our previous results.

Now we deal with the second part whether foreign equity investors in aggregate rebalance from the FX returns in EMs. Similarly, we estimate the following bivariate SVAR model of foreign equity flows and FX returns:

$$FXR_{i,t} = \alpha_{FXR} + \sum_{d=1}^5 \phi_{1,1,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * NF_{i,t-d} + u_{i,t}^{FXR} \quad (3a)$$

$$NF_{i,t} = \alpha_{NF} + \sum_{d=0}^5 \phi_{2,1,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * NF_{i,t-d} + u_{i,t}^{NF} \quad (3b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{FXR} \\ u_{i,t}^{NF} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{FXR,NF} \right], \mathbf{D}_i^{FXR,NF} = \begin{bmatrix} \sigma_{u,i,FXR}^2 & 0 \\ 0 & \sigma_{u,i,NF}^2 \end{bmatrix}$$

To make sure that the relationship between past FX returns and net flows is not just a proxy of the relationship between past equity local returns and net flows, we add five lags of equity local returns as an exogenous variable into the SVAR system as below:

$$FXR_{i,t} = \alpha_{FXR} + \sum_{d=1}^5 \phi_{1,1,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,3,d} * ELR_{i,t-d} + u_{i,t}^{FXR} \quad (4a)$$

$$NF_{i,t} = \alpha_{NF} + \sum_{d=0}^5 \phi_{2,1,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,3,d} * ELR_{i,t-d} + u_{i,t}^{NF} \quad (4b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{FXR} \\ u_{i,t}^{NF} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{FXR,NF} \right], \mathbf{D}_i^{FXR,NF} = \begin{bmatrix} \sigma_{u,i,FXR}^2 & 0 \\ 0 & \sigma_{u,i,NF}^2 \end{bmatrix}$$

Table III shows the results from the foreign equity flows equations for a bivariate SVAR of foreign equity flows and FX returns with no exogenous variable in Panel A, and with past equity local returns as an exogenous variable in Panel B, respectively. Like equity local returns, we also find a strong positive association between contemporaneous FX returns and foreign equity flows. However, the results about the effects of past FX returns on flows diverge in Panel A and B. Excluding equity local returns, both estimated coefficients and Granger causality tests presented in Panel A suggest that past FX returns positively predict flows over and above the predictions of lagged flows from five out of six markets except JSX. For instance, in Panel A of Table III, the estimated coefficients of one-day lagged FX returns for the five markets are strongly significant and ranging from 0.002 (Kospi, Kosdaq or PSE) to 0.006 (TWSE). Except JSX, the null hypothesis that Past FX returns do not Granger-cause flows are rejected in Granger causality test for other 5 markets at conventional 5% level (marginally rejected in TWSE).

However, this is no longer the case once we include equity local returns as an exogenous variable in our VAR framework. As it is shown in Panel B of Table III, once the equity local returns are included, both the strong evidence from the estimated coefficients of one-day lagged FX returns and the Granger tests become insignificant. For instance, the estimated coefficients

of one-day lagged FX returns becoming insignificant for all six markets in Panel B. Now the Granger tests can not reject the null hypothesis that past FX returns do not Granger-cause flows for all six markets at conventional level, while actually there is a substantial increase in every respective adjusted R^2 from Panel A to Panel B, invalidating the previous evidence in Panel A of Table III. Clearly, past FX returns only serve as a proxy capturing the effects of past equity local returns on flows when past equity local returns are excluded. Once equity local returns are included, there is little effect of past FX returns on flows, but only the effects of past equity local returns on flows. Figure II shows the responses of flows to a one-standard deviation innovation in FX returns with equity local returns in its 5 lags as an exogenous variable based on equation (4b) using GIRFs. We find that past FX movements have an insignificant influence on flows in most of the cases, which further supports our previous estimates.

Overall, we find the net flows of all foreign investors respond positively to past equity local returns in our sample of six EMs, which is in contrast to the evidence of portfolio-rebalancing by U.S. equity investors in Curcuro et al. (2014).¹⁵ But we find little evidence that foreign equity flows react to past currency movements, which is consistent with the very recent evidence of U.S. equity investors identified by Curcuro et al. (2014). The lack of sensitivity of foreign investors towards currency movements suggests that foreign equity investors in these six EMs only use FX as a vehicle (see, e.g., Goldberg and Tille, 2008; Devereux, and Shi, 2013).

4.2 The motives behind the responses: do foreign equity investors chase returns?

The above results are violating the first step and key driver of UEP (portfolio-rebalancing). As a result, it seems reasonable to attribute the failure of UEP in EMs (the positive correlation between equity local returns and FX returns in EMs) to the positive correlation between past equity local returns and foreign equity flows in EMs. Thus, as a natural next step, we ask the following questions: what is the exact motivation behind the positive responses of foreign equity investors to the shocks in past equity local returns? Is it return-chasing?

¹⁵ The difference between our results and the results of Curcuro et al. (2014) may be due to the fact that they only consider the US investors but we consider all foreign investors in aggregate. Even if US investors are somewhat informed or pursue portfolio-rebalancing strategy (as found in Curcuro, et al., 2011, 2014), the foreigners from the other countries other than the US may still pursue a return-chasing strategy because of less information, momentum, sentiment or other reasons.

Literature has offered two hypotheses to explain the positive correlation between past equity local returns and foreign equity flows. The first one is the so-called “return-chasing” hypothesis, which indicates that foreign investors react to past positive returns with positive inflows in order to chase high expected returns (Bohn and Tesar, 1996). This may arise because foreign investors extract information from past returns (Richards, 2005) if there is momentum in returns (Bekaert, et al., 2002).

The second hypothesis is what we term the macroeconomic news/sentiment hypothesis: Good (bad) news regarding the equity market leads to positive (negative) returns and to flows into (out of) equity markets (Ben-Rephael, et al., 2011).

We compare these two hypotheses in two ways. We first discuss the two hypotheses using the results we already obtained from the previous sub-section. If the return-chasing hypothesis dominates, there should be stronger forecasting power of the past equity local returns than the past FX returns on flows, as in the preliminary data analysis we find there is far less momentum in the FX returns which means that it is much more difficult for the investors to chase returns in the FX markets. If the macroeconomic news/sentiment hypothesis dominates, we conjecture that the past FX returns should have more forecasting power, as the FX markets are more liquid and easier to transmit news/sentiment (Bank for International Settlements, 2013). As we use the same flows series in equity and FX markets, the difference of results can only arise from the different properties of equity local returns and FX returns. The findings in the previous sub-section that the foreign equity flows respond to the past equity local returns (Table II) but not to the past FX returns (Table III), suggest a dominating role for the return-chasing hypothesis rather than the macroeconomic news/sentiment hypothesis.

After that, we further verify the return-chasing hypothesis by decomposing the current equity local returns (FX returns) into an expected component and an unexpected component, and checking the explanatory power of both components on current flows. If the return-chasing hypothesis is dominating, we should find the expected component to have a far better explanatory power on flows than the unexpected component. Otherwise, it means that the macroeconomic news/sentiment hypothesis is dominating. It is not uncommon to see that macroeconomic news or sentiments to be denoted by the unexpected components of returns in

literature. For instance, via a simple decomposition, Campbell (1991) show that the unexpected equity returns equal to cash flow news plus expected-return news. Similarly, Engle and Ng (1993) use an unexpected drop in returns as a proxy for bad news and an unexpected rise for good news. Although macroeconomic news or sentiment in returns may contain both an expected component and an unexpected component, it should be only the unexpected component which affects returns and flows (Ross et al., 1999).

We follow Richards (2005) and construct a series for “expected” returns on day t based on the return regressions in the VAR systems, using only variables predetermined up to day $t-1$, i.e., excluding same-day returns and flows, and a series for “unexpected” returns derived as actual returns less expected returns. The expected returns represent the information in past returns may be extracted by the foreign investors, and the unexpected returns represent the macroeconomic news or sentiment.¹⁶

To be specific, we use the following simple reduced-form VAR model with 5 lags to obtain expected equity local returns and unexpected equity local returns.

$$NF_{i,t} = \alpha_{NF} + \sum_{d=1}^5 \phi_{1,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * ELR_{i,t-d} + \varepsilon_{i,t}^{NF} \quad (5a)$$

$$ELR_{i,t} = \alpha_{ELR} + \sum_{d=1}^5 \phi_{2,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * ELR_{i,t-d} + \varepsilon_{i,t}^{ELR} \quad (5b)$$

where $\begin{bmatrix} \varepsilon_{i,t}^{NF} \\ \varepsilon_{i,t}^{ELR} \end{bmatrix} \approx N \left[0, \Sigma_i^{NF,ELR} \right], \Sigma_i^{NF,ELR} = \begin{bmatrix} \sigma_{\varepsilon,i,NF}^2 & \rho_{i,NF,ELR} \sigma_{\varepsilon,i,NF} \sigma_{\varepsilon,i,ELR} \\ \rho_{i,NF,ELR} \sigma_{\varepsilon,i,NF} \sigma_{\varepsilon,i,ELR} & \sigma_{\varepsilon,i,ELR}^2 \end{bmatrix}$

Similarly, we use the following simple reduced-form VAR model with 5 lags to obtain expected FX returns and unexpected FX returns.

$$NF_{i,t} = \alpha_{NF} + \sum_{d=1}^5 \phi_{1,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * FXR_{i,t-d} + \varepsilon_{i,t}^{NF} \quad (6a)$$

¹⁶ Our decomposition is plausible, but may not be the only way to distinguish the expected and the unexpected returns, or the return-chasing hypothesis and the macroeconomic news/sentiment hypothesis. We also decompose the returns by estimating a simple AR (1) model and take the predicted part as the expected returns and the residuals as unexpected returns, and get very similar results.

$$FXR_{i,t} = \alpha_{FXR} + \sum_{d=1}^5 \phi_{2,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * FXR_{i,t-d} + \varepsilon_{i,t}^{FXR} \quad (6b)$$

$$\text{where } \begin{bmatrix} \varepsilon_{i,t}^{NF} \\ \varepsilon_{i,t}^{FXR} \end{bmatrix} \approx N \left[0, \Sigma_i^{NF,FXR} \right], \Sigma_i^{NF,FXR} = \begin{bmatrix} \sigma_{\varepsilon,i,NF}^2 & \rho_{i,NF,FXR} \sigma_{\varepsilon,i,NF} \sigma_{\varepsilon,i,FXR} \\ \rho_{i,NF,FXR} \sigma_{\varepsilon,i,NF} \sigma_{\varepsilon,i,FXR} & \sigma_{\varepsilon,i,FXR}^2 \end{bmatrix}$$

After obtaining the expected and the unexpected equity local returns from model (5), we substitute the equity local returns in model (1) and (2) with its (un)expected component and re-estimate the model (1) and (2) to quantify the effects of its (un)expected equity local returns on foreign equity flows. The results are presented in Table IV. Similarly, we substitute the FX returns in model (3) and (4) with its expected and unexpected FX components obtained from model (6) to quantify the effects of the expected and the unexpected FX returns on foreign capital flows, with the results presented in Table V.

Table IV shows the results of explanatory power of the expected (unexpected) equity local returns on foreign equity flows in Panel A (Panel B), respectively. Clearly, flows are affected by both the expected and unexpected equity local returns according to the statistical significance of estimated coefficients of returns (the only exception is TWSE). More importantly, in terms of economic significance, in every case the magnitude of the effect from the expected component is much bigger than the one from the unexpected equity local returns. For instance, in the results from the simple bivariate VAR with no exogenous variable on the left-hand-side of Table IV, the estimated coefficients of the expected contemporaneous returns in Panel A range from 0.008 (PSE) to 0.172 (SET), roughly 10 times as much for the ones of the unexpected return in Panel B ranging from 0.001 (PSE) to 0.014 (TWSE), which attribute the major part of the effects of the past equity local returns on flows, to the expected returns rather than the unexpected returns associated with macroeconomic news or sentiment. When we include past FX returns as an exogenous variable, we find similar results which are reported on the right-hand-side of Table IV.

Similarly, Table V shows the results of explanatory power of expected (unexpected) FX returns on foreign equity flows in Panel A (Panel B), respectively. According to the statistical significance, flows are positively affected by the unexpected FX returns in every case, and the magnitude of coefficients is comparable to the ones of coefficients of unexpected equity local

returns in Table V. That is to say, flows are affected by unexpected FX returns as well as the unexpected equity local returns, although the magnitude of the effects from both of them is relatively small. It makes sense as the macroeconomic news/sentiment hypothesis should also work in FX markets if it works in the domestic equity markets. However, in stark contrast, we find little evidence that flows are affected by the expected FX returns (only in TWSE and SET). It may not be surprising to find that flows do not chase returns in FX markets given the fact that FX returns are notoriously difficult to forecast. Overall, we conclude the main explanation for the effects of past equity local returns on flows is the return-chasing hypothesis.¹⁷

4.3 The second step of UEP: do foreign equity flows cause FX returns?

Preliminary data analysis also suggests a positive contemporaneous correlation between foreign equity flows and FX returns in our sample of six EMs. However, it is unclear whether it is only a contemporaneous relationship, or an inter-temporal relationship (e.g., foreign equity flows predict FX returns over and above the predictions of lagged FX returns). Using monthly data, Curcuru et al. (2014) “do not know the timing of purchases within a month and so cannot perfectly disentangle” a contemporaneous relationship from an inter-temporal relationship, as the inter-temporal changes in variables within the month are netted out. Our data set in daily frequency is in a much better position regarding this question.

To examine the second step of UEP whether foreign equity flows cause FX returns in the same direction in EMs, we estimate the following bivariate SVAR, which is in line with the previous literature on flows and exchange rates such as Hau and Rey (2004), Froot and Ramadorai (2005) and Love and Payne (2008), and focus on the FX returns equation from now onwards:

$$NF_{i,t} = \alpha_{NF} + \sum_{d=1}^5 \phi_{1,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * FXR_{i,t-d} + u_{i,t}^{NF} \quad (7a)$$

¹⁷ We try to replicate Table 4 in page 10 of Richards (2005) but include both realized and implied global equity volatility and FX volatility obtained from hourly data. While we find that both the past global equity and FX volatility have a negative effect on flows into EMs, the explanatory power of past global equity/FX volatilities cannot be beat past equity local returns. Consistent with Richards (2005), we find the best explanatory power comes from previous day's Nasdaq returns for Kosdaq and TWSE, previous day's Philadelphia Semiconductor Index returns for Kospi, domestic returns for the rest three markets, which suggests foreign investors also extract information from the markets outside of the EMs.

$$FXR_{i,t} = \alpha_{FXR} + \sum_{d=0}^5 \phi_{2,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * FXR_{i,t-d} + u_{i,t}^{FXR} \quad (7b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{NF} \\ u_{i,t}^{FXR} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{NF,FXR} \right], \mathbf{D}_i^{NF,FXR} = \begin{bmatrix} \sigma_{u,i,NF}^2 & 0 \\ 0 & \sigma_{u,i,FXR}^2 \end{bmatrix}$$

To make sure that the relationship between net flows and FX returns is not just a proxy of the relationship between net flows and equity local returns, we add five lags of equity local returns as an exogenous variable into the SVAR system as below:

$$NF_{i,t} = \alpha_{NF} + \sum_{d=1}^5 \phi_{1,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{1,2,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{1,3,d} * ELR_{i,t-d} + u_{i,t}^{NF} \quad (8a)$$

$$FXR_{i,t} = \alpha_{FXR} + \sum_{d=0}^5 \phi_{2,1,d} * NF_{i,t-d} + \sum_{d=1}^5 \phi_{2,2,d} * FXR_{i,t-d} + \sum_{d=1}^5 \phi_{2,3,d} * ELR_{i,t-d} + u_{i,t}^{FXR} \quad (8b)$$

$$\text{where } \begin{bmatrix} u_{i,t}^{NF} \\ u_{i,t}^{FXR} \end{bmatrix} \approx N \left[0, \mathbf{D}_i^{NF,FXR} \right], \mathbf{D}_i^{NF,FXR} = \begin{bmatrix} \sigma_{u,i,NF}^2 & 0 \\ 0 & \sigma_{u,i,FXR}^2 \end{bmatrix}$$

The results for the FX returns equation from the bivariate SVAR without exogenous variable and with equity local returns as an exogenous variable are shown in Panel A and B of Table IV, respectively. We report the estimates of the contemporaneous net flows and the past net flows up to 5 lags, adjusted R^2 and the p-value of Granger causality test for the hypothesis: Past flows do not Granger-cause FX returns. In general, the results from Panel A and B are similar. From estimated coefficients of flows, we can see that flows have an overall positive influence on FX returns, as the sum of the coefficients of both contemporaneous and past foreign equity flows are always positive. However, most of the positive influence comes from the contemporaneous flows, with the estimated coefficient ranging from 1.831 (SET) to 3.908 (JSX) in Panel A, and from 1.688 (SET) to 3.811 (Kospi). There are slight reversals as there are some negative coefficients of past flows sometimes. The adjusted R^2 ranges from 0.009 (PSE) to 0.116 (TWSE) in Panel B, which is no less to the explanatory power of currency order flows in the existing literature (e.g., from 0.0036 (GBP/EUR) to 0.006 (USD/EUR) in Table 5 of Love and Payne, 2008). The Granger-causality tests reject the null hypothesis that past flows do not Granger-cause current FX returns for half of our sample countries at 15% level

(Indonesia, Taiwan and Thailand)¹⁸. In other words, we find a strong contemporaneous positive relationship between foreign equity flows and FX returns, and a weak inter-temporal relationship which implies that foreign equity flows may also have a positive and permanent impact on future FX returns.

Figure III presents the responses of FX returns to a one-standard-deviation innovation in foreign equity flows using general impulse response functions (GIRFs). We only report the results based on equation (8b), as the results based on equation (8a) are similar. Except Indonesia (JSX), we cannot find substantial reversals in the other 5 EMs. We find that the responses of FX returns become insignificant from the next trading day for Indonesia (JSX), Korea (Kospi), Korea (Kosdaq) and Philippines (PSE) but last for almost one week for Taiwan (TWSE) and Thailand (SET). More important, over a 10-day period, we find that the median cumulative response on FX returns of a one-standard-deviation shock in foreign net flows is 3.86% based on equation (7b) and 4.16% based on equation (8b), which means that it is not only statistically significant but also economically significant. Overall, our results strongly support the second step of UEP: foreign equity flows have a strong positive and significant influence on FX returns.

5. Comparison with other explanations

In this section, we discuss other explanations of the failure of UEP proposed in the existing studies and compare them with our return-chasing explanation. To the best of our knowledge, we are only aware of the following two other explanations¹⁹: 1) market risk (Kim, 2011), and 2) flight-to-quality (Cho et al., 2014).

Using data for 4 EMs (Singapore, Korea, Malaysia and Thailand), Kim (2011) argue that the positive correlation between FX and equity local returns in EMs might be explained by market risks in EMs, due to “incomplete institutional reforms, weaker macroeconomic fundamentals, more volatile economic conditions, shallow financial markets and imperfect

¹⁸ The weak evidence may be due to the information loss in net equity flows. Compared to order flows, net equity flows have no information about the signs of the trade, i.e., the initiated side of the trades.

¹⁹ In the latest version, Cenedese et al. (2014) find that global equity volatility risk can only partially explain the cross-sectional failure of UEP, which motivates our further robustness tests controlling for VIX and/or other global shocks.

market integration”. We hold the belief that market risk may not be enough to fully explain the failure of UEP in our sample of six EMs, for the following three reasons.

First of all, it is less intuitive how can market risks affect the sign of the corrections between FX and equity local returns. It is more likely that market risks affect the magnitude rather than the sign of the correlations, such as in the case of Cho et al. (2014).

Moreover, we find an obvious upward time trend in the 250-trading-day (one calendar year) moving correlations between equity local returns and FX returns in six EMs in Figure IV, while Kim (2011, p1492) suggests that the magnitude of the correlation coefficients should have decreased, as “the market risk after the liberalization of financial markets is expected to decrease gradually along the path of market integration”. Appendix A shows that the upward trend does not change qualitatively no matter we calculate 125-trading-day (half a year), 63-trading-day (one calendar quarter) or 21-trading-day (one calendar month) moving correlations. However, the upward supports our return-chasing explanation as it has become increasingly safer and easier for the foreign investors to chase returns along the path of market integration.

Finally, Kim (2011) suggests that the magnitude of the correlations in relatively more developed EMs (Singapore and Korea), which is generally associated with less market risks, should be smaller than the ones in relatively less developed EMs (Malaysia and Thailand). However, over the full sample we find in the preliminary analysis that the magnitude of the correlation coefficients in the relatively more developed EMs such as Kospi, Kosdaq and TWSE (0.30, 0.29 and 0.31, respectively) are larger than in the relatively less developed EMs such as JSX, PSE and SET (0.20, 0.24 and 0.20, respectively)²⁰. Again, this fact supports our return-chasing explanation as the relatively more developed EMs are more attractive to the foreign investors in terms of chasing returns.

Cho et al. (2014) propose the flight-to-quality phenomenon as an explanation of the positive correlation between quarterly FX and equity local returns, as “When we partition the sample into up markets and down markets, we find that net capital flows are sensitive to overall

²⁰ In fact, Richards (2005, p5) documents that the annual turnover ratio (the annual turnover divided by the previous day’s market capitalization) in Kospi, Kosdaq and TWSE (is 2.32, 9.85 and 2.08, respectively), is much higher than JSX, PSE and SET (0.38, 0.07 and 1.05, respectively), while the same ratio is only 0.89 for New York Stock Exchange in 2001. We confirm this result using recent data.

stock market conditions only in down markets, consistent with the flight-to-quality arguments”.

In order to distinguish from flight-to-quality, we follow Cho et al. (2014) and partition our sample of daily data into global up (when the returns of MSCI World index is positive) and down (when the returns of MSCI World index is negative) periods but find both positive correlations between flows and local equity returns, and between FX and equity local returns in global up periods as well as in global down periods in the four two columns of Panel A of Table VII. Alternatively, we redefine global up markets as the period when the local equity returns and the returns of MSCI EM index is positive and global down markets as the period when the local equity returns or the returns of MSCI EM index is negative, and find similar results in the later four columns of Panel A of Table VII.

In order to not shatter the continuity of our sample, in Panel B of Table VII, we further divide our sample into an earlier Asian Financial Crisis and Dotcom Crisis subsample (from various starting date to Oct 9, 2002), a non-crisis subsample (from Oct 9, 2002 to Aug 9, 2007), and a recent Global Financial Crisis subsample (from Aug 9, 2007 to Dec 30, 2013) and find similar results. As a result, we conclude that our results do not fully rely on flight-to-quality and our explanation applies in general.

6. Robustness tests

In this section, we discuss the robustness of our analysis. In particular, we consider the following four possible concerns: 1) the model reliability, 2) using return differentials, 3) the changes in financial wealth, and 4) the regional co-movement effect.

One concern of our analysis is the reliability of our estimates. We perform the following robustness tests: using flows without winsorization, or using 1-day lagged flow data. We also introduce S&P 500 returns, or Nasdaq returns, or Philadelphia Semiconductor index returns as in Richards (2005) once a time as control variables, as both the two markets in Korea and the Taiwan market are technology-intensive and highly depends on the global technology cycle. We also introduce proxies for global developed market information (MSCI World index returns), global emerging market information (MSCI EM index returns) and global risk appetite (VIX) as Ulku and Weber (2014) into our VAR models once a time as control variables. As

shown from Panel A to Panel H in Appendices from B to H, all key results stay, essentially, unchanged. Our results are also robust to various combinations of different control variables, tri-variate VAR models, alternative order of variables and alternative number of lags, but we do not report the results for space constraints.

Another concern arises because we use raw returns and Hau and Rey (2006) use return differentials between U.S. and foreign stock indices. While Hau and Rey (2006) build their theory in a world with two countries and an exogenous setting of portfolio-rebalancing regarding return differentials, there are more than two countries in this real world and it is not straight forward which country should be used as the benchmark (Richards, 2005, p8), especially when we focus on all the foreign investors rather than only the investors from the U.S.²¹ As it is shown in Panel C of Table VII, we find positive and significant correlations between flows and equity returns differentials, and between FX returns and equity return differentials in most of the cases when we use different benchmarks (S&P 500, Nasdaq, Philadelphia Semiconductor index, MSCI world index, MSCI EM index) to construct equity returns differentials. In particular, we find three negative and significant correlations between FX returns and equity returns differentials when we use MSCI EM index to construct equity returns differentials, which is probably due to some kind of portfolio-rebalancing regarding the MSCI EM returns.²²

Since we use flow data rather than portfolio data, like most of the literature, our analysis is also subject to the critique from Curcuru et al. (2011): flow data are influenced by changes in financial wealth. Like most of the literature about the interaction between international capital flows and domestic equity returns (Froot et al., 2001; Griffin et al., 2004; Richard, 2005), we scaled our flow data by local equity market capitalization in our main analysis. Alternatively, we also try to control for the changes in financial wealth of investors by normalizing our flows variable by trading volume instead of local equity market capitalization, or scaling flows by

²¹ For instance, Kim (2011) finds significant different results using Japan rather than the U.S. as a benchmark economy. Cho et al. (2014) also find significant different results once Japan is included.

²² For instance, Richards (2005, p8) explicitly points out: “Much investment in emerging markets occurs not via managers with a global mandate but rather via specialist managers investing only in emerging markets. Hence if portfolio rebalancing effects are important, the relevant return might not be a global mature markets return, but rather the return on a basket of emerging market equities.” See also the relevant references cited by Richards (2005).

the average of absolute flows of previous 21/63/125/250 trading days. The key results stay unchanged. As shown from Panel I to Panel L in Appendices from 3B to 3H, all key results stay unchanged.

Since our six sample countries are geographically close, there might be a common regional effect behind the flows and returns. In unreported results, we have found strong co-movements in flows, FX returns and equity local returns, with the average correlation coefficients between net flows into different markets of approximately 0.21, while the average correlation coefficients of FX returns and equity local returns between different markets are 0.33 and 0.47, respectively. In unreported principal component analysis, we find that the first principal component is able to explain 36%, 47% and 56% of the variations in net flows, FX returns and equity local returns, respectively, which suggests that there are regional/global co-movements within flows, FX returns and equity local returns. We take the co-movements into account by employing the fixed-effect panel-VAR regression²³. Generally, all previous results are confirmed by the panel-VAR approach. In Figure V, we find foreign equity flows have significant positive influence on FX returns, and equity local returns have a significant positive influence on future foreign equity flows. However, the influence of equity local returns on future FX returns, and the influence of FX returns on future foreign equity flows are insignificant.

7. Conclusion

Within the portfolio-rebalancing framework, UEP suggests that the equity local returns and FX returns are negatively correlated. Motivated by the failure of UEP in EMs, this paper examines the mechanisms underlying UEP as well as its prediction, using daily foreign equity flows, equity local returns and FX returns data for six Asian markets from the 1990s to 2013. Previous literature either only investigates the mechanisms underlying UEP (Curcuru et al., 2014), or only examines its prediction (Kim, 2011; Cho et al., 2014; Cenedese et al., 2014). We find evidence unsupportive of some mechanisms underlying UEP. For example, we find little

²³ As there are many more observations over time than across countries in our study, we prefer using a fixed-effect panel-VAR regression to Arellano-Bond estimation in the dynamic panel.

evidence that foreign equity investors in aggregate react to currency movements, suggesting that foreign investors in aggregate in EMs mainly use exchange rates as a vehicle (e.g., Goldberg and Tile, 2008; Devereux, and Shi, 2013), which is consistent with the evidence found in Curcuru et al. (2014) for U.S. equity investors. Furthermore, foreign equity investors in aggregate pursue a return-chasing rather than portfolio-rebalancing strategy regarding equity local returns in EMs, which leads to a positive correlation between the equity local returns and FX returns in EMs. However, we do find strong support for the rest of UEP: we find a strong contemporaneous positive relationship between foreign equity flows and FX returns, and a weak inter-temporal relationship which implies that foreign equity flows may also have a positive and permanent impact on future FX returns.

The main contribution of this paper is that we unveil another side of the UEP: UEP does not automatically guarantee a negative correlation between equity local returns and FX returns. Instead, UEP provides one explanation of the relationship but may not be the only one. The relationship between FX and equity local returns hinges on the overall behavior of foreign equity investors in aggregate, i.e., whether they pursue a portfolio-rebalancing or return-chasing strategy. The former predicts a negative correlation, while the latter creates a positive correlation between FX and equity local returns. In other words, we reconcile the mixed evidence on the prediction of UEP about the correlation between FX and equity local returns in previous literature. By doing so, we contribute to the notoriously difficult question on the prediction of FX movements.

Our results are complementary rather than contradictory to the previous literature on UEP. In particular, we find that foreign equity investors in aggregate pursue a return-chasing strategy, but we do not exclude the possibility that a specific group of investors such as the U.S. equity investors might pursue a portfolio-rebalancing strategy regarding equity local returns in EMs, which is of course possible, so long as they form a small part of the total number of foreign investors. It is also possible that the foreign equity investors pursue a portfolio-rebalancing strategy regarding equity local returns in developed markets, which we are refrained from commenting due to data limitation.

The findings of this paper have important implications for policy-makers, academics and

investors. For policy-makers in EMs, they should not only pay attention to the equity markets when there are net foreign equity flows, but also to the FX markets as foreign equity flows have a positive influence on FX markets as well. For academics, other missing rings of UEP call for research. For instance, there is a possibility that equity local returns and FX returns are uncorrelated, if all foreign investors in aggregate do not react to the movement in equity local returns. For hedgers, our results suggest that the FX hedging strategy might be helpful to foreigners' equity investments in EMs, as FX movements do not offset equity local return but add additional risks to the total portfolio returns in EMs. For speculators and arbitrageurs, trading strategies based on UEP (e.g., Cenedese et al., 2014) can be improved by incorporating international capital flows.

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Table I. Descriptive statistics

This table provides descriptive statistics of the three main economic variables, i.e.: net flows (NF_{it}), equity local returns (ELR_{it}) and FX returns (FXR_{it}) in daily frequency for six equity markets from various starting dates to the end of 2013. Net flows are defined as (buy value) - (sell value) by foreign investors, scaled by previous day's market capitalization. To control for abnormal capital inflows, observations above the 99th percentile of the daily net flow distribution are set equal to the 99th percentile point. Equity prices are expressed in local currency and both equity local returns and FX returns are in percentage terms. For each country the table shows the starting date of the sample, the mean, median, and standard deviation of net flows and returns, the first five autocorrelation estimates for each series, and the contemporaneous correlation between net flows and equity local returns, equity local returns and FX returns, and net flows and FX returns, respectively. The end date for all countries is December 30, 2013. * and bold mean that the correlation coefficient is significant at the 5% level or better here.

Country	Start Date		Mean	Median	Std.Dev.	AC (1)	AC (2)	AC (3)	AC (4)	AC (5)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*FXR)
Indonesia (JSX)	Sept. 9, 1996	NF	0.006	0.002	0.026	0.18*	0.11	0.09	0.09	0.06	0.29*	0.20*	0.05*
		ELR	0.049	0.100	1.696	0.14*	0.02	-0.03	-0.02	-0.02			
		FXR	-0.039	0.000	1.724	-0.02	0.08	-0.01	-0.03	-0.03			
Korea (Kospi)	Jun. 30, 1997	NF	0.004	0.001	0.039	0.48*	0.32*	0.26*	0.23*	0.22*	0.31*	0.30*	0.11*
		ELR	0.024	0.086	1.942	0.06*	-0.04	-0.02	-0.04	-0.04			
		FXR	-0.004	0.022	1.054	0.02	-0.10*	-0.01	-0.07*	-0.11*			
Korea (Kosdaq)	Mar. 15, 1999	NF	0.003	0.001	0.029	0.42*	0.26*	0.22*	0.22*	0.20*	0.19*	0.29*	0.08*
		ELR	-0.013	0.130	2.053	0.14*	0.04	0.03	0.02	-0.02			
		FXR	0.004	0.026	0.716	-0.02	0.03	-0.04	0.02	-0.03			
Philippines (PSE)	Mar. 15, 1999	NF	0.001	0.000	0.013	0.17*	0.14*	0.11*	0.10*	0.08*	0.17*	0.24*	0.06*
		ELR	0.031	0.035	1.384	0.12*	0.00	-0.04	-0.02	-0.04			
		FXR	-0.004	0.000	0.446	-0.03	-0.04	0.03	-0.04	-0.01			
Taiwan (TWSE)	Jan. 1, 2001	NF	0.006	0.006	0.040	0.51*	0.33*	0.26*	0.22*	0.18*	0.51*	0.31*	0.32*
		ELR	0.019	0.053	1.413	0.05*	0.02	0.01	-0.01	-0.02			
		FXR	0.003	0.000	0.265	0.03	0.02	-0.01	0.02	0.06*			
Thailand (SET)	Dec. 1, 1997	NF	0.001	0.000	0.030	0.56*	0.38*	0.29*	0.25*	0.21*	0.37*	0.20*	0.13*
		ELR	0.031	0.034	1.641	0.07*	0.05*	0.00	-0.01	0.00			
		FXR	0.006	0.000	0.529	0.12*	-0.03	-0.05*	0.02	0.11*			

Table II. Explanatory power of equity local returns on foreign equity flows

This table shows the results from the flows equations for a bivariate structural vector autoregression (SVAR) of foreign equity flows and equity local returns (ELR) with no exogenous variable in Panel A, and with FX returns as an exogenous variable in Panel B, respectively. The VAR is structural as we include contemporaneous equity local returns in the flows equation. The SVARs are estimated separately for each country with five lags. The L is the lag operator. We report t -statistics computed using maximum likelihood estimates of covariance matrix in parentheses, while *, ** and *** indicates statistical significance at the 10%, 5% and 1% level, respectively. We report adjusted R^2 and the p -values of Granger causality test for the hypothesis: Past equity local returns do not Granger-cause flows.

	Panel A: Without exogenous variable						Panel B: With past FX returns as an exogenous variable					
	JSX	Kospi	Kosdaq	PSE	TWSE	SET	JSX	Kospi	Kosdaq	PSE	TWSE	SET
EquityLocalReturns	0.004*** (18.88)	0.005*** (21.91)	0.002*** (10.13)	0.001*** (10.03)	0.014*** (38.73)	0.006*** (26.12)	0.004*** (18.83)	0.005*** (21.98)	0.002*** (10.23)	0.001*** (9.91)	0.014*** (38.69)	0.006*** (25.96)
L.EquityLocalReturns	0.002*** (8.25)	0.005*** (18.00)	0.002*** (8.44)	0.001*** (7.60)	0.004*** (8.39)	0.004*** (19.09)	0.002*** (8.34)	0.005*** (17.92)	0.002*** (8.16)	0.001*** (7.55)	0.004*** (8.31)	0.004*** (18.60)
L2.EquityLocalReturns	0.001** (2.20)	-0.001*** (-3.91)	-0.001*** (-6.33)	0.001*** (3.48)	0.001 (1.37)	-0.001*** (-3.53)	0.001** (2.52)	-0.001*** (-3.46)	-0.001*** (-5.38)	0.001*** (3.43)	0.001 (1.40)	-0.001*** (-3.59)
L3.EquityLocalReturns	0.000 (1.23)	-0.000 (-0.96)	-0.001*** (-2.99)	0.000*** (3.22)	0.001 (1.18)	-0.001*** (-2.99)	0.000 (1.39)	-0.000 (-0.56)	-0.001*** (-3.13)	0.000*** (3.10)	0.001 (1.60)	-0.001*** (-2.91)
L4.EquityLocalReturns	-0.001** (-2.42)	-0.000 (-1.15)	-0.000 (-0.45)	0.000 (1.56)	-0.000 (-0.75)	-0.001*** (-4.20)	-0.000** (-2.07)	-0.000 (-1.06)	-0.000 (-0.11)	0.000* (1.79)	-0.000 (-0.86)	-0.001*** (-3.96)
L5.EquityLocalReturns	-0.000 (-0.33)	-0.000 (-1.45)	-0.000 (-1.01)	0.000 (0.61)	-0.000 (-1.15)	-0.001*** (-3.39)	-0.000 (-0.38)	-0.000* (-1.73)	-0.000 (-1.07)	0.000 (0.68)	-0.001 (-1.30)	-0.001*** (-3.13)
Adj. R2	0.149	0.391	0.252	0.105	0.516	0.479	0.149	0.391	0.253	0.104	0.516	0.479
Granger	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table III. Explanatory power of FX returns on foreign equity flows

This table shows the results from the flows equations for a bivariate structural vector autoregression (SVAR) of foreign equity flows and FX returns (FXR) with no exogenous variable in Panel A, and with equity local returns as an exogenous variable in Panel B, respectively. The VAR is structural as we include contemporaneous FX returns in the flows equation. Both SVARs are estimated separately for each country with five lags. The *L*. is the lag operator. We report *t*-statistics computed using maximum likelihood estimates of covariance matrix in parentheses, while *, ** and *** indicates statistical significance at the 10%, 5% and 1% level, respectively. We report adjusted R² and the *p*-values of Granger causality test for the hypothesis: Past FX returns do not Granger-cause flows.

	Panel A: Without exogenous variable						Panel B: With past equity local returns as an exogenous variable					
	JSX	Kospi	Kosdaq	PSE	TWSE	SET	JSX	Kospi	Kosdaq	PSE	TWSE	SET
FXReturns	0.001*** (3.75)	0.004*** (6.97)	0.003*** (4.59)	0.002*** (3.76)	0.042*** (19.45)	0.004*** (5.47)	0.001*** (2.83)	0.004*** (7.61)	0.003*** (4.96)	0.002*** (3.85)	0.042*** (19.64)	0.004*** (4.83)
L.FXReturns	0.000 (0.61)	0.002*** (4.78)	0.002*** (3.22)	0.002*** (3.46)	0.006** (2.45)	0.003*** (4.32)	-0.000 (-1.64)	-0.000 (-0.76)	0.000 (0.31)	0.001 (1.41)	0.003 (1.38)	0.001* (1.79)
L2.FXReturns	-0.000 (-1.56)	0.000 (0.97)	-0.002*** (-2.85)	0.001 (1.50)	0.002 (0.97)	-0.000 (-0.47)	-0.000* (-1.95)	0.001 (1.12)	-0.001** (-1.99)	-0.000 (-0.94)	0.001 (0.55)	-0.000 (-0.31)
L3.FXReturns	-0.000 (-0.21)	-0.001 (-1.45)	-0.000 (-0.21)	0.001* (1.86)	-0.004* (-1.72)	-0.001 (-1.20)	-0.000 (-0.05)	-0.001* (-1.79)	0.000 (0.58)	0.000 (0.54)	-0.005** (-2.03)	0.000 (0.06)
L4.FXReturns	-0.000 (-1.33)	-0.001 (-1.31)	-0.001 (-1.12)	0.000 (0.43)	0.000 (0.16)	-0.000 (-0.03)	-0.000 (-1.02)	0.000 (0.10)	-0.001 (-0.83)	-0.000 (-0.52)	0.000 (0.03)	0.000 (0.14)
L5.FXReturns	-0.000 (-0.45)	-0.000 (-0.01)	-0.000 (-0.26)	0.000 (0.21)	-0.001 (-0.30)	-0.002*** (-2.66)	-0.000 (-0.44)	0.000 (0.36)	0.000 (0.23)	-0.000 (-0.34)	-0.000 (-0.09)	-0.002** (-2.50)
Adj. R2	0.052	0.271	0.212	0.061	0.356	0.339	0.079	0.328	0.236	0.084	0.367	0.393
Granger	0.392	0.000	0.001	0.005	0.073	0.000	0.168	0.391	0.371	0.577	0.273	0.096

Table IV. Explanatory power of expected and unexpected equity local returns on foreign equity flows

This table shows the results from the flows equations for a bivariate structural vector autoregression (SVAR) of foreign equity flows and expected (unexpected) equity local returns without (with) FX returns as an exogenous variable in Panel A (Panel B). The VAR is structural as we include contemporaneous expected (unexpected) equity local returns in the flows equation. The SVAR is estimated separately for each country with five lags. The L is the lag operator. We report maximum likelihood estimates of t -statistics in parentheses, while *, ** and *** indicates statistical significance at the 10%, 5% and 1% level, respectively. We report adjusted R^2 and the p -values of Granger causality test for the hypothesis: Past expected (unexpected) equity local returns do not Granger-cause flows.

	Without exogenous variable						With past FX returns as an exogenous variable					
	JSX	Kospi	Kosdaq	PSE	TWSE	SET	JSX	Kospi	Kosdaq	PSE	TWSE	SET
Panel A: Flows equations with expected equity local returns												
ExpectedEquityLocalReturns	0.017***	0.036***	0.016***	0.008***	0.041*	0.172***	0.018***	0.036***	0.016***	0.008***	0.041*	0.167***
	(10.83)	(13.81)	(9.72)	(7.32)	(1.68)	(18.99)	(10.92)	(13.05)	(9.25)	(6.79)	(1.66)	(18.27)
L.ExpectedEquityLocalReturns	0.004**	-0.001	-0.011***	0.004***	0.005	-0.056***	0.004***	-0.002	-0.010***	0.004***	0.007	-0.055***
	(2.27)	(-0.51)	(-6.34)	(3.20)	(0.21)	(-6.06)	(2.61)	(-0.84)	(-5.51)	(3.09)	(0.27)	(-5.97)
L2.ExpectedEquityLocalReturns	0.006***	-0.002	-0.007***	0.004***	0.021	0.052***	0.006***	-0.002	-0.007***	0.004***	0.021	0.050***
	(3.41)	(-0.79)	(-4.00)	(4.02)	(1.20)	(5.36)	(3.49)	(-0.64)	(-4.09)	(3.63)	(1.19)	(5.14)
L3.ExpectedEquityLocalReturns	-0.001	0.000	-0.001	0.003***	-0.017	0.005	0.000	0.000	-0.001	0.003***	-0.015	0.004
	(-0.35)	(0.17)	(-0.79)	(2.59)	(-1.43)	(0.56)	(0.03)	(0.09)	(-0.57)	(2.60)	(-1.33)	(0.40)
L4.ExpectedEquityLocalReturns	0.004***	0.009***	0.003*	0.003***	0.006	-0.020***	0.005***	0.009***	0.003	0.003***	0.005	-0.019***
	(2.73)	(3.58)	(1.89)	(2.99)	(0.58)	(-3.69)	(2.79)	(3.56)	(1.63)	(2.77)	(0.48)	(-3.47)
L5.ExpectedEquityLocalReturns	-0.001	0.004*	-0.001	0.003**	0.007	0.006**	-0.001	0.005**	-0.001	0.003**	0.007	0.006**
	(-0.54)	(1.79)	(-0.74)	(2.39)	(0.84)	(2.34)	(-0.60)	(2.18)	(-0.65)	(2.44)	(0.81)	(2.32)
Adj. R2	0.079	0.292	0.231	0.075	0.279	0.389	0.079	0.280	0.231	0.068	0.280	0.356
Granger	0.000	0.002	0.000	0.000	0.548	0.000	0.000	0.020	0.000	0.000	0.487	0.421
Panel B: Flows equations with unexpected equity local returns												
UnexpectedEquityLocalReturns	0.004***	0.005***	0.002***	0.001***	0.014***	0.006***	0.004***	0.005***	0.002***	0.001***	0.014***	0.006***
	(18.89)	(21.91)	(10.15)	(9.99)	(38.79)	(26.77)	(18.84)	(21.99)	(10.25)	(9.87)	(38.76)	(26.62)
L.UnexpectedEquityLocalReturns	0.003***	0.005***	0.002***	0.001***	0.004***	0.005***	0.003***	0.005***	0.002***	0.001***	0.004***	0.005***
	(10.93)	(18.88)	(9.81)	(8.85)	(8.46)	(20.42)	(10.95)	(18.71)	(9.47)	(8.70)	(8.38)	(19.79)
L2.UnexpectedEquityLocalReturns	0.001***	-0.001***	-0.001***	0.001***	0.001	-0.001***	0.001***	-0.001***	-0.001***	0.001***	0.001	-0.001***
	(3.69)	(-4.75)	(-5.00)	(4.29)	(1.25)	(-2.92)	(3.99)	(-4.28)	(-4.09)	(4.20)	(1.28)	(-3.07)
L3.UnexpectedEquityLocalReturns	0.000	-0.001***	-0.001***	0.000***	0.000	-0.001***	0.000	-0.001***	-0.001***	0.000***	0.001	-0.001***
	(1.06)	(-3.45)	(-3.40)	(2.97)	(0.92)	(-3.52)	(1.27)	(-3.01)	(-3.42)	(2.83)	(1.34)	(-3.39)
L4.UnexpectedEquityLocalReturns	-0.001***	-0.001***	-0.000	0.000	-0.001	-0.001***	-0.001***	-0.001***	-0.000	0.000	-0.001	-0.001***
	(-3.16)	(-2.82)	(-0.60)	(1.29)	(-1.18)	(-4.46)	(-2.77)	(-2.73)	(-0.26)	(1.48)	(-1.26)	(-4.29)
L5.UnexpectedEquityLocalReturns	-0.000	-0.001***	-0.000	-0.000	-0.001*	-0.001***	-0.000	-0.001***	-0.000	-0.000	-0.001**	-0.001***
	(-1.45)	(-3.40)	(-1.39)	(-0.30)	(-1.91)	(-3.61)	(-1.40)	(-3.68)	(-1.37)	(-0.21)	(-2.06)	(-3.47)
Adj. R2	0.149	0.391	0.252	0.104	0.516	0.484	0.150	0.391	0.253	0.104	0.517	0.483
Granger	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table V. Explanatory power of expected and unexpected FX returns on foreign equity flows

This table shows the results from the flows equations for a bivariate structural vector autoregression (SVAR) of foreign equity flows and expected (unexpected) FX returns without (with) equity local returns as an exogenous variable in Panel A (Panel B). The VAR is structural as we include contemporaneous expected (unexpected) FX returns in the flows equation. The SVAR is estimated separately for each country with five lags. The L is the lag operator. We report t -statistics computed using maximum likelihood estimates of covariance matrix in parentheses, while *, ** and *** indicates statistical significance at the 10%, 5% and 1% level, respectively. We report adjusted R^2 and the p -values of Granger causality test for the hypothesis: Past expected (unexpected) FX returns do not Granger-cause flows.

	Without exogenous variable						With past equity local returns as an exogenous variable					
	JSX	Kospi	Kosdaq	PSE	TWSE	SET	JSX	Kospi	Kosdaq	PSE	TWSE	SET
Panel A: Flows equations with expected FX returns												
ExpectedFXReturns	-0.002	0.002	-0.020	-0.018***	0.083**	0.020***	-0.002	-0.001	0.005	-0.003	0.089**	0.009*
	(-0.96)	(0.65)	(-1.22)	(-3.05)	(2.25)	(3.99)	(-0.98)	(-0.34)	(0.31)	(-0.43)	(2.39)	(1.88)
L.ExpectedFXReturns	-0.000	0.006*	0.008	-0.010*	-0.094**	-0.007	-0.000	0.004	0.014	-0.000	-0.092**	-0.007
	(-0.11)	(1.75)	(0.48)	(-1.73)	(-2.55)	(-1.38)	(-0.16)	(1.29)	(0.80)	(-0.00)	(-2.51)	(-1.51)
L2.ExpectedFXReturns	-0.005**	0.003	-0.005	0.006	0.034	-0.003	-0.005**	-0.000	-0.029	0.008	0.036	-0.003
	(-2.20)	(0.97)	(-0.27)	(1.04)	(0.88)	(-0.67)	(-2.16)	(-0.07)	(-1.64)	(1.41)	(0.94)	(-0.64)
L3.ExpectedFXReturns	-0.002	0.000	-0.013	0.023***	-0.067*	0.003	-0.002	0.003	-0.023	0.019***	-0.070*	0.002
	(-0.74)	(0.14)	(-0.84)	(4.02)	(-1.83)	(0.65)	(-0.91)	(1.12)	(-1.57)	(3.28)	(-1.93)	(0.45)
L4.ExpectedFXReturns	0.000	-0.001	0.017*	0.005	-0.012	-0.018***	0.000	-0.003	0.013	0.003	-0.011	-0.015***
	(0.02)	(-0.38)	(1.77)	(0.98)	(-0.34)	(-3.59)	(0.09)	(-0.93)	(1.34)	(0.54)	(-0.32)	(-3.10)
L5.ExpectedFXReturns	-0.001	0.004	0.018**	-0.002	0.025	0.004	0.001	0.007**	0.014	-0.002	0.018	0.005
	(-0.42)	(1.44)	(2.09)	(-0.34)	(1.06)	(0.85)	(0.37)	(2.27)	(1.60)	(-0.38)	(0.74)	(1.20)
Adj. R2	0.049	0.258	0.206	0.059	0.280	0.331	0.076	0.320	0.232	0.083	0.291	0.390
Granger	0.298	0.089	0.174	0.001	0.106	0.013	0.874	0.125	0.195	0.015	0.137	0.113
Panel B: Flows equations with unexpected FX returns												
UnexpectedFXReturns	0.001***	0.004***	0.003***	0.002***	0.042***	0.005***	0.001***	0.004***	0.003***	0.002***	0.042***	0.004***
	(3.74)	(7.02)	(4.58)	(3.74)	(19.50)	(6.02)	(2.83)	(7.69)	(4.96)	(3.84)	(19.68)	(5.56)
L.UnexpectedFXReturns	0.000	0.002***	0.002***	0.002***	0.006***	0.005***	-0.000*	-0.000	0.000	0.001	0.004*	0.002***
	(0.52)	(4.74)	(3.11)	(3.27)	(2.72)	(5.93)	(-1.69)	(-0.75)	(0.21)	(1.21)	(1.66)	(3.22)
L2.UnexpectedFXReturns	-0.000	-0.000	-0.002***	0.000	0.002	0.000	-0.000	-0.000	-0.001*	-0.001	0.001	0.000
	(-1.13)	(-0.20)	(-2.71)	(1.05)	(0.93)	(0.61)	(-1.51)	(-0.12)	(-1.78)	(-1.35)	(0.48)	(0.43)
L3.UnexpectedFXReturns	-0.000	-0.001**	-0.000	0.001	-0.005**	-0.001*	-0.000	-0.001**	0.000	0.000	-0.005**	-0.000
	(-0.18)	(-2.56)	(-0.24)	(1.41)	(-2.10)	(-1.66)	(-0.13)	(-2.31)	(0.44)	(0.27)	(-2.39)	(-0.35)
L4.UnexpectedFXReturns	-0.000*	-0.001**	-0.001	0.000	0.001	0.000	-0.000	-0.000	-0.001	-0.000	0.001	0.000
	(-1.67)	(-2.52)	(-1.33)	(0.09)	(0.46)	(0.23)	(-1.33)	(-0.96)	(-0.88)	(-0.75)	(0.36)	(0.58)
L5.UnexpectedFXReturns	-0.000	-0.001	-0.000	-0.000	0.002	-0.001	-0.000	-0.000	0.000	-0.000	0.002	-0.001
	(-0.57)	(-1.52)	(-0.33)	(-0.21)	(0.84)	(-1.49)	(-0.58)	(-0.72)	(0.07)	(-0.61)	(1.03)	(-1.52)
Adj. R2	0.052	0.271	0.212	0.061	0.356	0.339	0.080	0.329	0.236	0.084	0.367	0.395
Granger	0.453	0.000	0.002	0.017	0.018	0.000	0.166	0.170	0.492	0.498	0.082	0.027

Table VI. The price impact of foreign equity flows on FX returns

This table shows the results from the FX returns equations for a bivariate structural vector autoregression (SVAR) of foreign equity flows and FX returns with no exogenous variable in Panel A, and with equity local returns as an exogenous variable in Panel B, respectively. The VAR is structural as we include contemporaneous flows in the FX returns equation. Both SVARs are estimated separately for each country with five lags. The L is the lag operator. We report t -statistics computed using maximum likelihood estimates of covariance matrix in parentheses, while *, ** and *** indicates statistical significance at the 10%, 5% and 1% level, respectively. We report adjusted R^2 and the p -values of Granger causality test for the hypothesis: Past flows do not Granger-cause FX returns.

	Panel A: Without exogenous variables						Panel B: With past equity local returns as an exogenous variable					
	JSX	Kospi	Kosdaq	PSE	TWSE	SET	JSX	Kospi	Kosdaq	PSE	TWSE	SET
Flows	3.908***	3.361***	2.087***	2.258***	2.507***	1.831***	2.985***	3.811***	2.290***	2.345***	2.550***	1.688***
	(3.75)	(6.97)	(4.59)	(3.76)	(19.45)	(5.47)	(2.83)	(7.61)	(4.96)	(3.85)	(19.64)	(4.83)
L.Flows	0.604	-0.404	0.242	-0.264	-0.579***	1.005***	-0.643	-0.453	0.196	-0.111	-0.367**	0.755*
	(0.57)	(-0.78)	(0.50)	(-0.43)	(-3.93)	(2.69)	(-0.58)	(-0.82)	(0.40)	(-0.18)	(-2.20)	(1.88)
L2.Flows	1.120	0.611	0.147	0.064	0.058	-0.193	1.025	0.058	-0.100	0.221	0.090	-0.284
	(1.06)	(1.17)	(0.31)	(0.10)	(0.39)	(-0.52)	(0.93)	(0.10)	(-0.20)	(0.36)	(0.53)	(-0.71)
L3.Flows	2.367**	0.086	0.982**	0.318	-0.057	-0.772**	3.204***	0.769	1.040**	0.350	-0.025	-0.858**
	(2.24)	(0.17)	(2.04)	(0.52)	(-0.39)	(-2.06)	(2.91)	(1.39)	(2.12)	(0.57)	(-0.15)	(-2.13)
L4.Flows	-4.144***	0.534	-0.710	0.540	-0.327**	0.129	-2.337**	0.650	-0.505	0.552	-0.392**	0.325
	(-3.92)	(1.03)	(-1.47)	(0.89)	(-2.22)	(0.34)	(-2.12)	(1.18)	(-1.03)	(0.90)	(-2.35)	(0.81)
L5.Flows	-0.084	-0.089	-0.212	0.895	-0.078	-0.029	0.311	-0.542	-0.124	0.943	-0.118	0.175
	(-0.08)	(-0.18)	(-0.47)	(1.49)	(-0.58)	(-0.09)	(0.29)	(-1.09)	(-0.27)	(1.55)	(-0.78)	(0.51)
Adj. R2	0.015	0.046	0.011	0.009	0.115	0.049	0.024	0.052	0.014	0.009	0.116	0.049
Granger	0.001	0.567	0.246	0.502	0.000	0.040	0.021	0.383	0.389	0.444	0.003	0.133

Table VII. Robustness checks on correlations

This table shows the robustness results on the correlations between our three key variables, i.e.: foreign net flows (NF_{it}), FX returns (FXR_{it}) and equity local returns (ELR_{it}) in daily frequency for six equity markets. Panel A shows correlations between FX returns and equity local returns during global up and down periods according to different definitions. Panel B shows the correlations between FX returns and equity local returns in crisis and non-crisis subsamples. Panel C shows the correlations between FX returns and equity return differentials using different benchmarks. Stars mean that the correlation coefficient is significant at the 5% level or better.

Panel A: Correlations between FX returns and equity local returns during global up and down periods												
	Up period (when the returns of MSCI world index are positive, replicating Cho et al., 2014)		Down period (when the returns of MSCI world index are negative, replicating Cho et al., 2014)		Up period (when equity local returns are positive)		Down period (when equity local returns are negative)		Up period (when the returns of MSCI EM index are positive)		Down period (when the returns of MSCI EM index are negative)	
	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*ELR)	corr(ELR*FXR)
Indonesia (JSX)	0.3109*	0.2039*	0.2725*	0.2088*	0.2344*	0.1988*	0.1894*	0.2365*	0.3157*	0.1179*	0.2713*	0.1472*
Korea (Kospi)	0.2583*	0.2258*	0.2506*	0.2707*	0.2185*	0.2203*	0.2024*	0.2782*	0.2383*	0.1604*	0.2223*	0.2513*
Korea (Kosdaq)	0.0980*	0.2273*	0.2196*	0.2783*	0.0943*	0.1737*	0.0648*	0.2538*	0.1534*	0.1083*	0.2067*	0.2809*
Philippines (PSE)	0.1606*	0.3401*	0.2328*	0.2383*	0.1147*	0.1843*	0.1686*	0.1890*	0.1599*	0.2404*	0.2149*	0.1830*
Taiwan (TWSE)	0.3701*	0.2530*	0.5381*	0.2688*	0.2230*	0.1694*	0.4687*	0.1879*	0.3427*	0.1981*	0.4855*	0.2263*
Thailand (SET)	0.3340*	0.1584*	0.3741*	0.2017*	0.2485*	0.2006*	0.2869*	0.1437*	0.3567*	0.1489*	0.2911*	0.1294*

Panel B: Correlations between FX returns and equity local returns during Asian Financial Crisis and Dotcom crisis, non-crisis and Global Financial Crisis periods									
	Asian and Dotcom Crisis (before Oct 9, 2002)			Non-crisis (from Oct 9, 2002 to Aug 9, 2007)			Global Financial Crisis (after Aug 9, 2007)		
	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*FXR)	corr(NF*ELR)	corr(ELR*FXR)	corr(NF*FXR)
Indonesia (JSX)	0.3239*	0.1832*	0.0692*	0.2688*	0.3872*	0.0719*	0.3005*	0.3460*	0.0928*
Korea (Kospi)	0.3132*	0.2331*	0.0647*	0.2336*	0.1810*	0.1147*	0.3962*	0.5346*	0.2463*
Korea (Kosdaq)	0.1396*	0.1811*	0.0936*	0.2234*	0.1667*	0.0406	0.3086*	0.4882*	0.1457*
Philippines (PSE)	0.2847*	0.2104*	0.0737*	0.1492*	0.1832*	0.0719*	0.1365*	0.3114*	0.0527*
Taiwan (TWSE)	0.3922*	0.1378*	0.1427*	0.4851*	0.2472*	0.3209*	0.6007*	0.4143*	0.4207*
Thailand (SET)	0.3947*	0.2148*	0.1144*	0.3912*	0.1235*	0.1886*	0.3609*	0.2806*	0.1723*

Panel C: Correlations between FX returns and equity return differentials (ΔELR) with different benchmarks										
	S&P500		Nasdaq		Philadelphia Semiconductor index		MSCI World		MSCI EM	
	corr(NF*ΔELR)	corr(ΔELR*FXR)	corr(NF*ΔELR)	corr(ΔELR*FXR)	corr(NF*ΔELR)	corr(ΔELR*FXR)	corr(NF*ΔELR)	corr(ΔELR*FXR)	corr(NF*ΔELR)	corr(ΔELR*FXR)
Indonesia (JSX)	0.2343*	0.1254*	0.2026*	0.1001*	0.1546*	0.0628*	0.2476*	0.1282*	0.1848*	0.0597*
Korea (Kospi)	0.2690*	0.2073*	0.2278*	0.1731*	0.1615*	0.1217*	0.2607*	0.1942*	0.1493*	0.0703*
Korea (Kosdaq)	0.1617*	0.1599*	0.1302*	0.1254*	0.0968*	0.0867*	0.1531*	0.1178*	0.1086*	-0.0590*
Philippines (PSE)	0.1292*	0.1139*	0.1136*	0.0866*	0.0925*	0.0494*	0.1363*	0.0585*	0.0977*	-0.1085*
Taiwan (TWSE)	0.3682*	0.1712*	0.3053*	0.1490*	0.2023*	0.1014*	0.3414*	0.1241*	0.1295*	-0.0836*
Thailand (SET)	0.3047*	0.1429*	0.2599*	0.1101*	0.1896*	0.0574*	0.2938*	0.1136*	0.1447*	0.0149

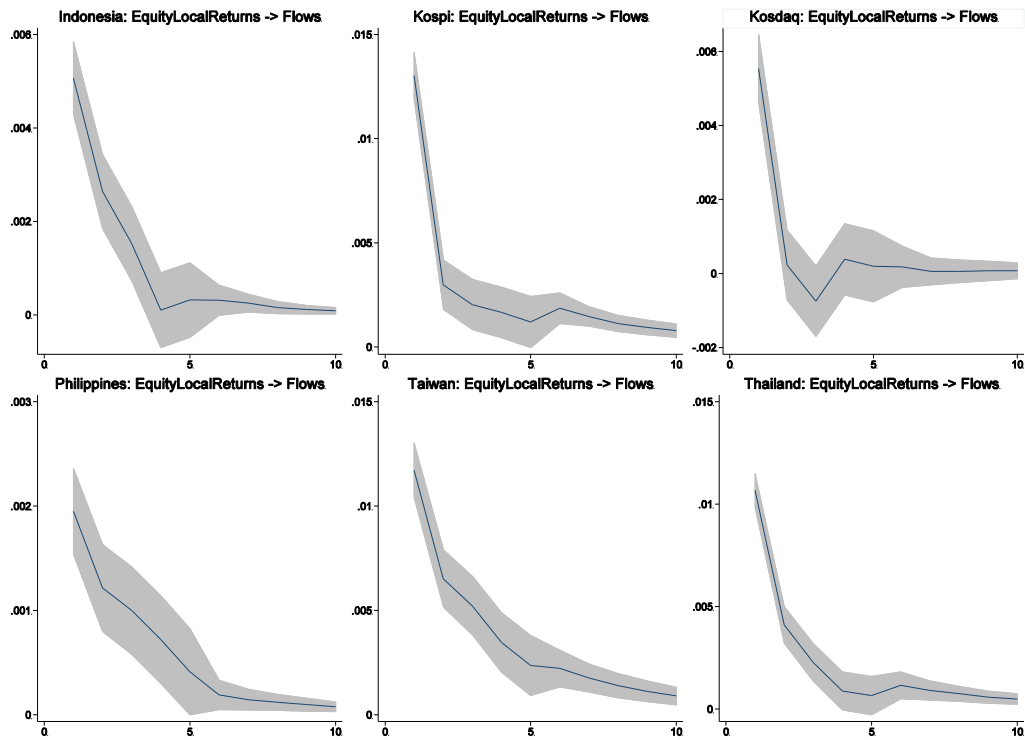


Figure I. Responses of foreign equity flows to equity local returns shocks.

This figure shows the responses of foreign equity flows to a one-standard-deviation innovation in equity local returns using general impulse response function from the next trading day. The estimates are obtained from a structural bivariate vector autoregressive models (SVAR) of foreign equity flows and equity local returns with FX returns in its past 5 lags as an exogenous variable. The VAR is estimated for each market separately with 5 lags, using daily data from various starting dates to the end of 2013. The grey area is 95% confidence intervals based on asymptotic standard errors.

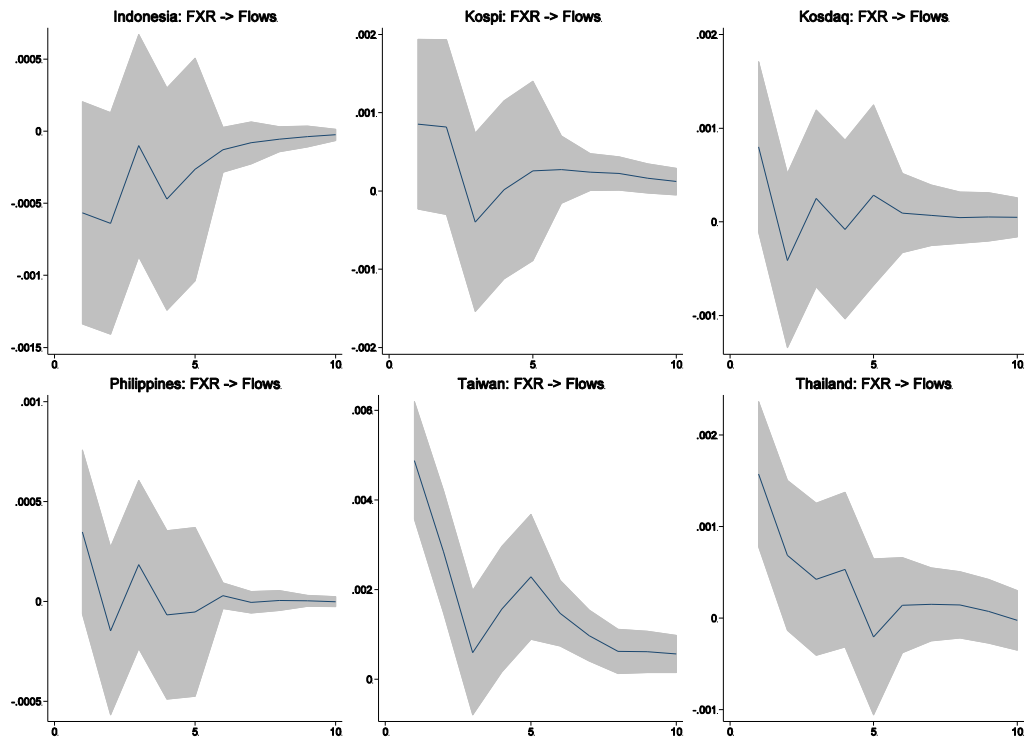


Figure II. Responses of foreign equity flows to FX shocks.

This figure shows the responses of foreign equity flows to a one-standard-deviation innovation in FX returns using general impulse response function from the next trading day. The estimates are obtained from a structural bivariate vector autoregressive models (SVAR) of foreign equity flows and FX returns with equity local returns in its past 5 lags as an exogenous variable. The VAR is estimated for each market separately with 5 lags, using daily data from various starting dates to the end of 2013. The grey area is 95% confidence intervals based on asymptotic standard errors.

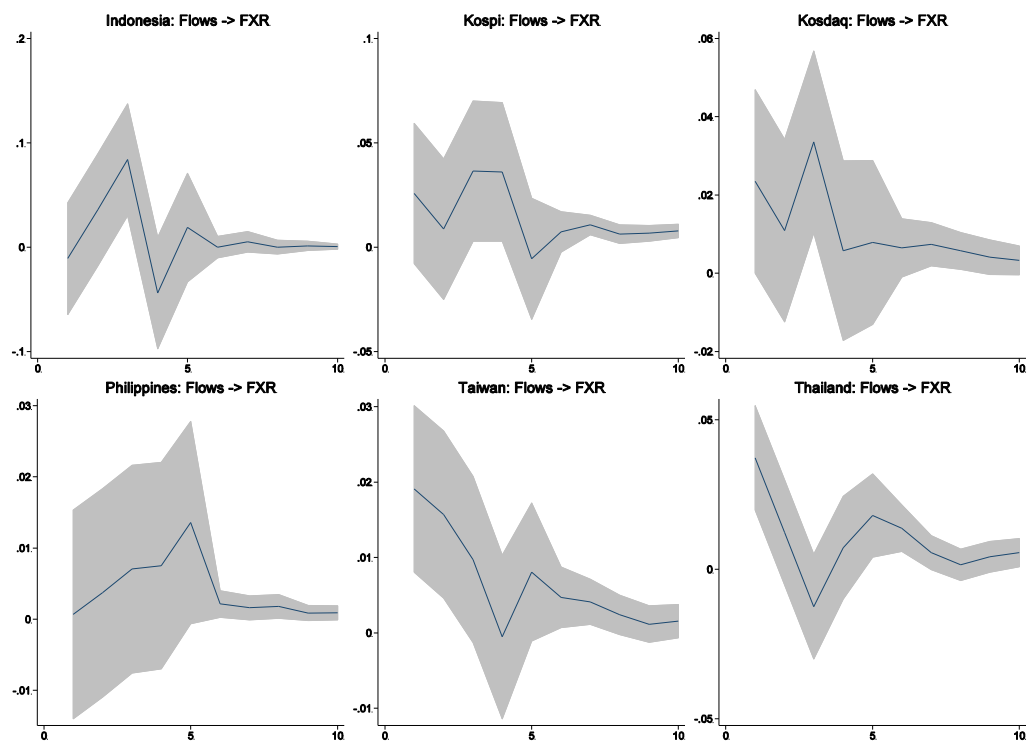


Figure III. Responses of FX returns to foreign equity flows shocks.

This figure shows the responses of FX returns (FXR) to a one-standard-deviation innovation in foreign equity flows using general impulse response function from the next trading day. The estimates are obtained from a structural bivariate vector autoregressive models (SVAR) of foreign equity flows and FX returns with equity local returns in its past 5 lags as an exogenous variable. The VAR is estimated for each market separately with 5 lags, using daily data from various starting dates to the end of 2013. The grey area is 95% confidence intervals based on asymptotic standard errors.

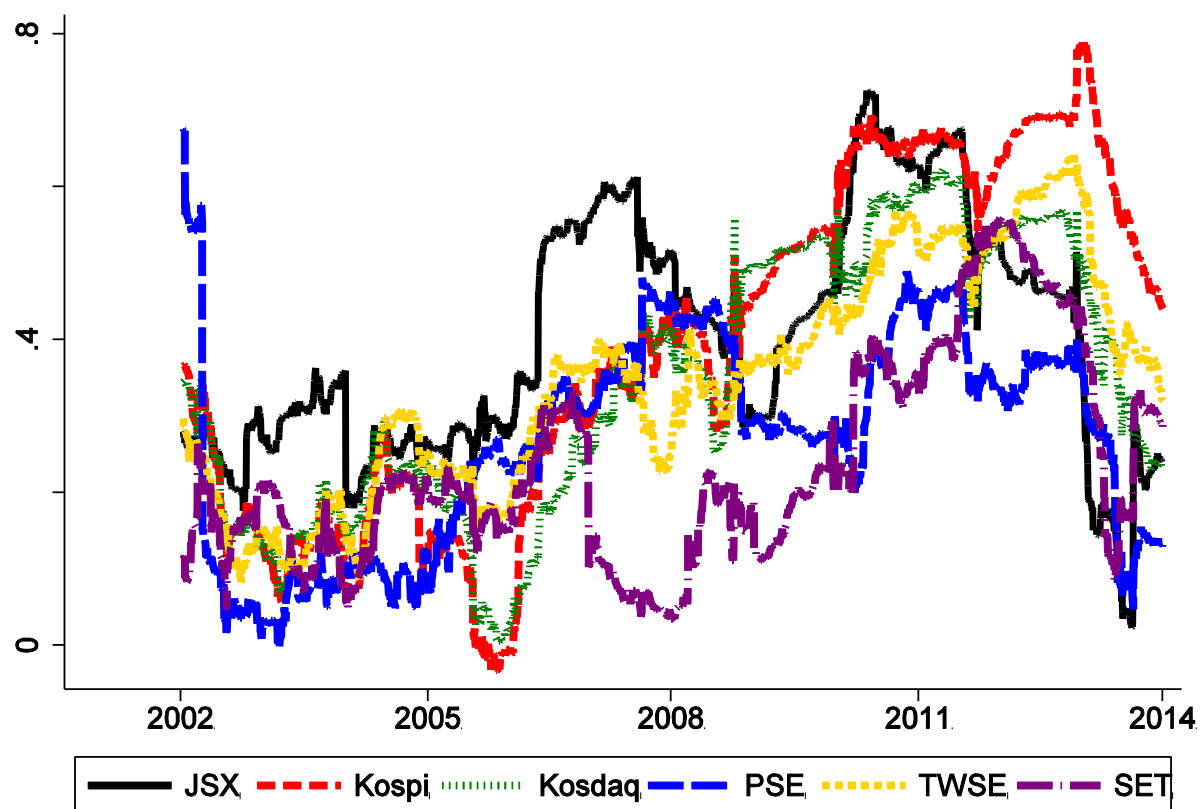


Figure IV. Moving correlation between equity local returns and FX returns.

This figure plots the 250-trading-day moving (rolling) correlations between equity local returns and FX returns for the six emerging markets in our sample.

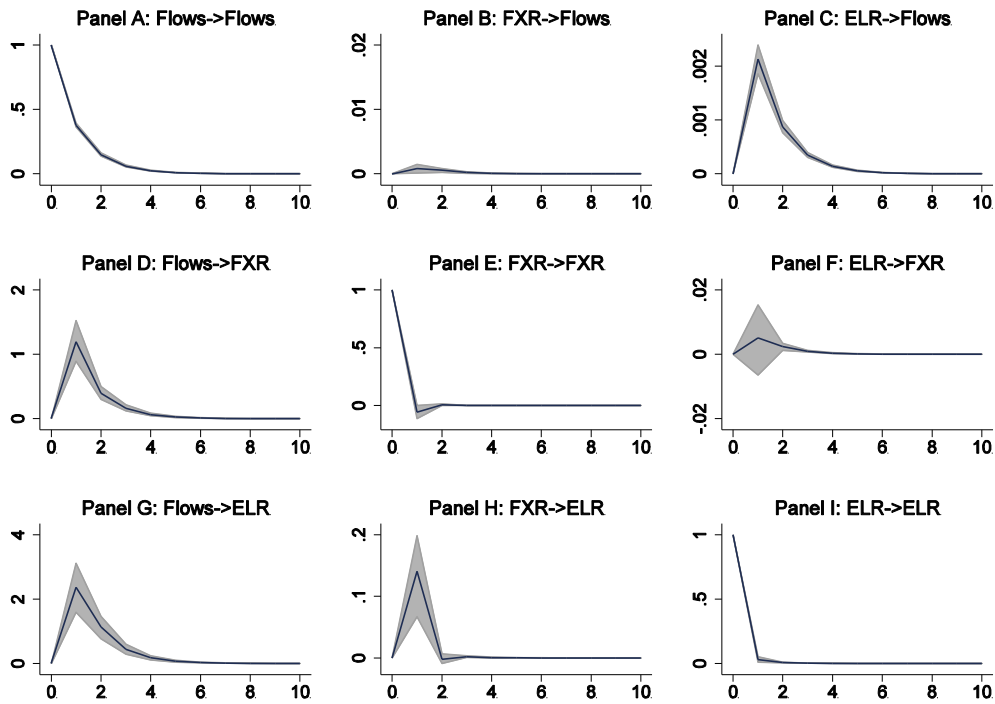


Figure V. Impulse responses analyses of a tri-variate panel-VAR system.

This figure shows the responses of flows, FX returns (FXR) and equity local returns (ELR) to a one-standard-deviation innovation in flows in the left-top, left-middle and left-bottom panels, respectively; the responses of flows, FX returns and equity local returns to a one-standard-deviation innovation in FX returns in the middle-top, middle-middle and middle-bottom panels, respectively; and the responses of flows, FX returns and equity local returns to a one-standard-deviation innovation in equity local returns in the right-top, right-middle and right-bottom panels, respectively. The estimates are obtained from a fixed-effect tri-variate panel-VAR with 5 lags using daily data from January 1, 2001 (due to the data availability of Taiwan) to the end of 2013. The grey area is 95% confidence intervals based on asymptotic standard errors.